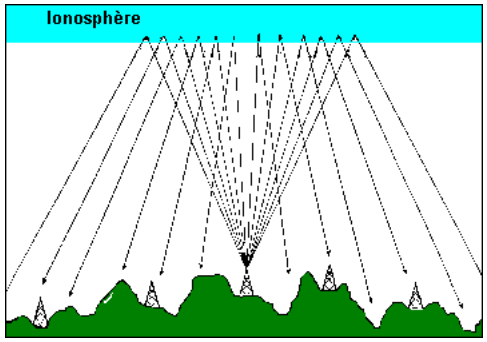


Reliable State-Wide HF Communications

Lew Thompson

W5IFQ

3 NOV 2026

Introduction

- Reliable state and FEMA Region 6 HF Communications can be achieved using the correct type of antenna and selecting the optimum operating frequency.
- This type of HF operation is referred to as NVIS (Near Vertical Incident Skywave) communications.
- Region 6 Army MARS has 6 NVIS nets a week (mornings and evening) and successfully moves digital and SSB traffic across Texas, Louisiana, Arkansas and New Mexico every net.

Goal

- Introduce Near Vertical Incident Sky-Wave (NVIS) propagation theory and practice.

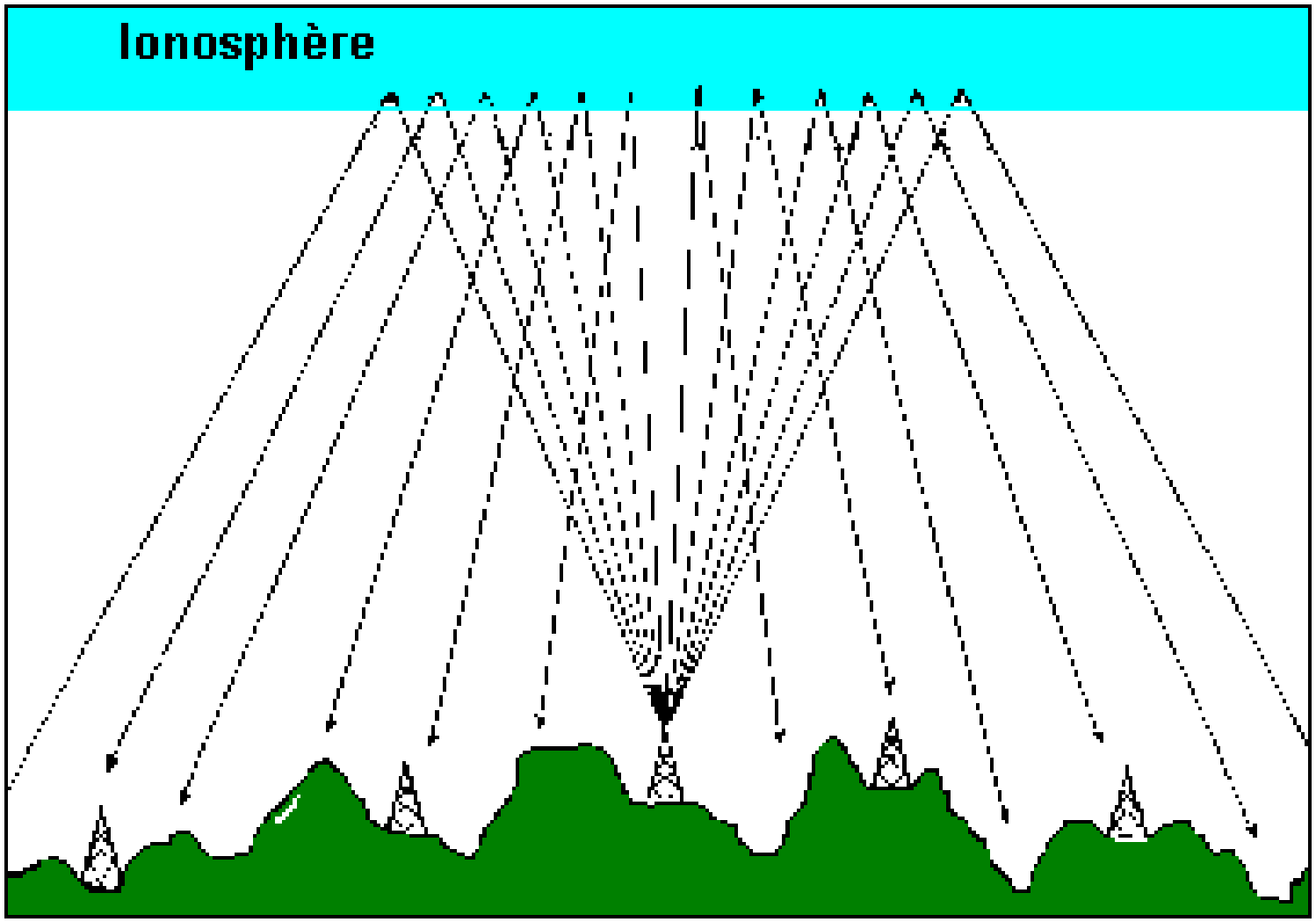
Outline

- HF Propagation Modes
- NVIS hardware and modes
  - Antenna Requirements
  - Equipment Requirements
  - Potential operating modes
- Antenna Types
  - Low Dipoles
  - Long-Wire
  - Loops

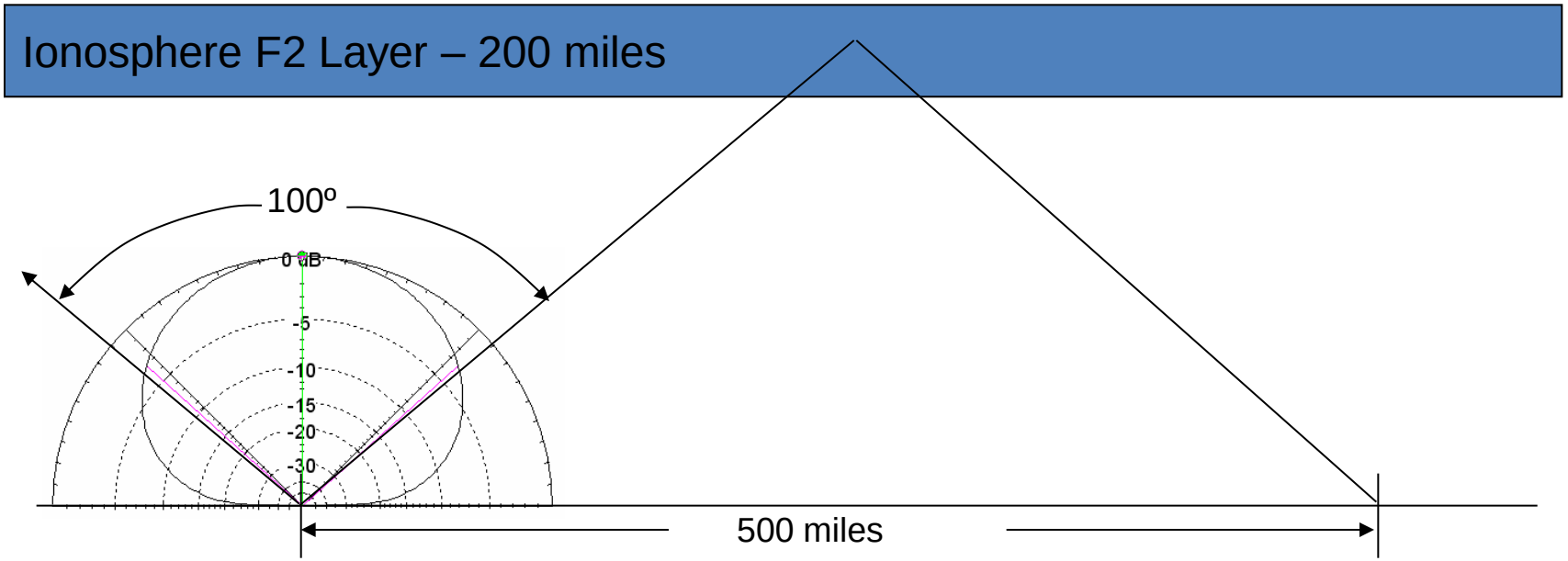
HF Propagation Modes
(3 – 30 MHz)

- Free Space – Line of sight
- Ground Wave – Follows Earth's curvature
- Ionospheric Skip
  - Long Distance with a "skip-zone"
  - NVIS (Near Vertical Incidence Sky-Wave)

NVIS Propagation

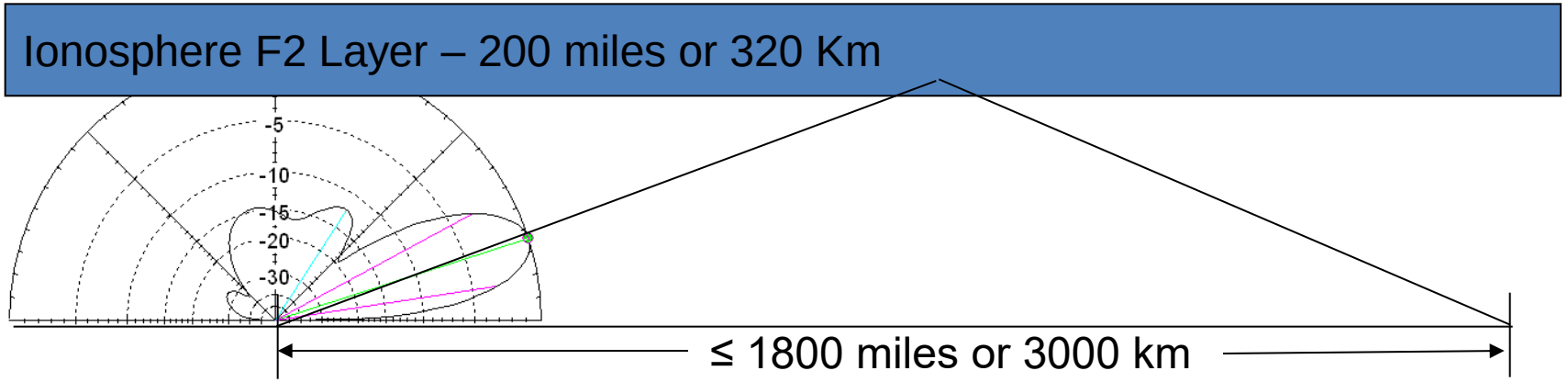


Required NVIS Antenna Pattern (Cloud Warmer)



- Want to optimize take-off angles from 50º to 90º
- Elevation Beam width = 100º
- Centered on the vertical axis.

IN CONTRAST - TYPICAL LONG-RANGE PROPAGATION



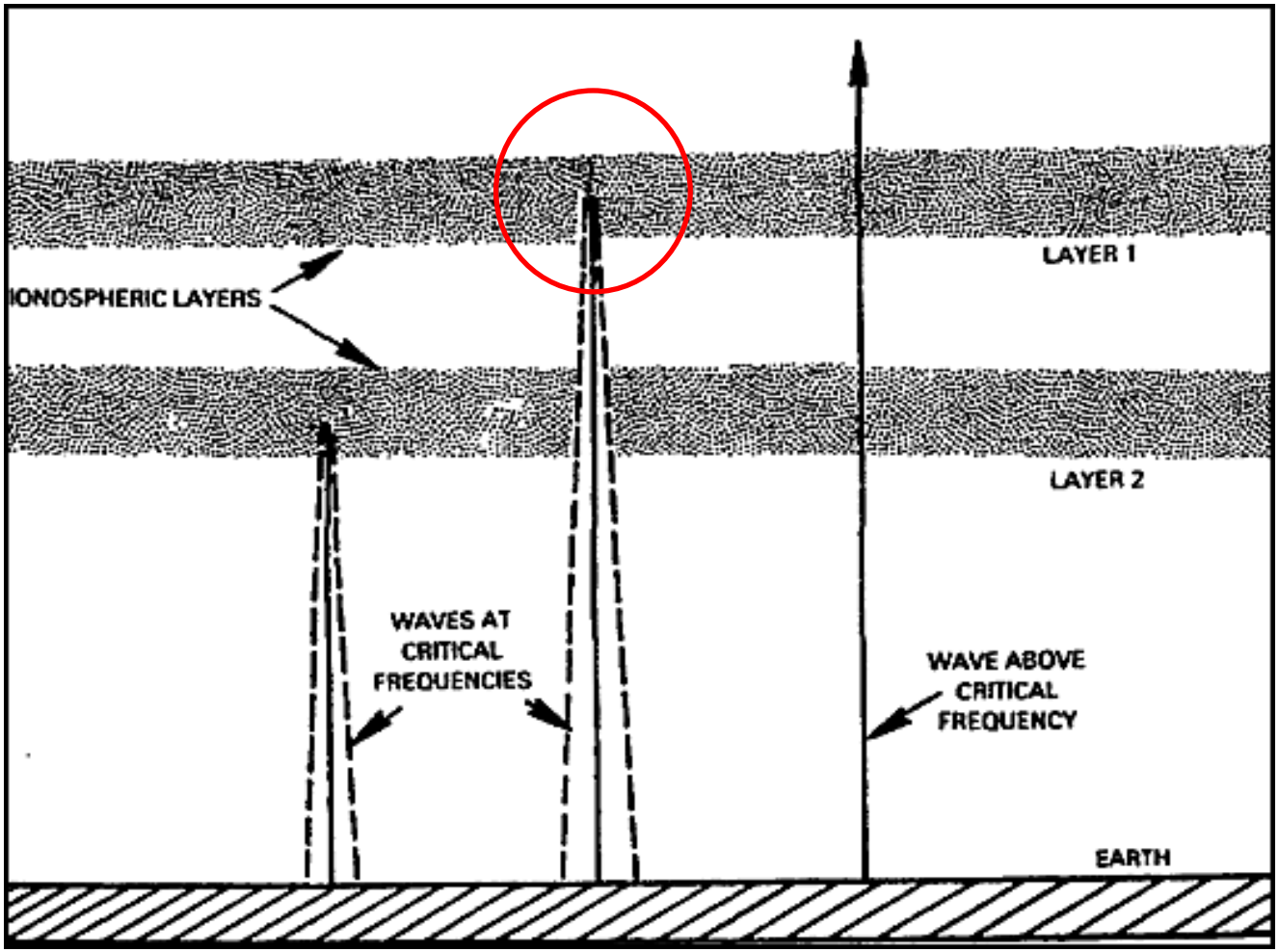
Both F2 & E layers propagation can be involved in multiple reflection circuits.

NVIS PARAMETERS

- Operating Frequency
- Antenna design
  - Correct directivity pattern
  - Provide usable impedance to RF transmitter

Critical Frequency Definition

- Must operate at or below the local Critical Frequency (CF) in the F2 layer.



F2 layer

E layer

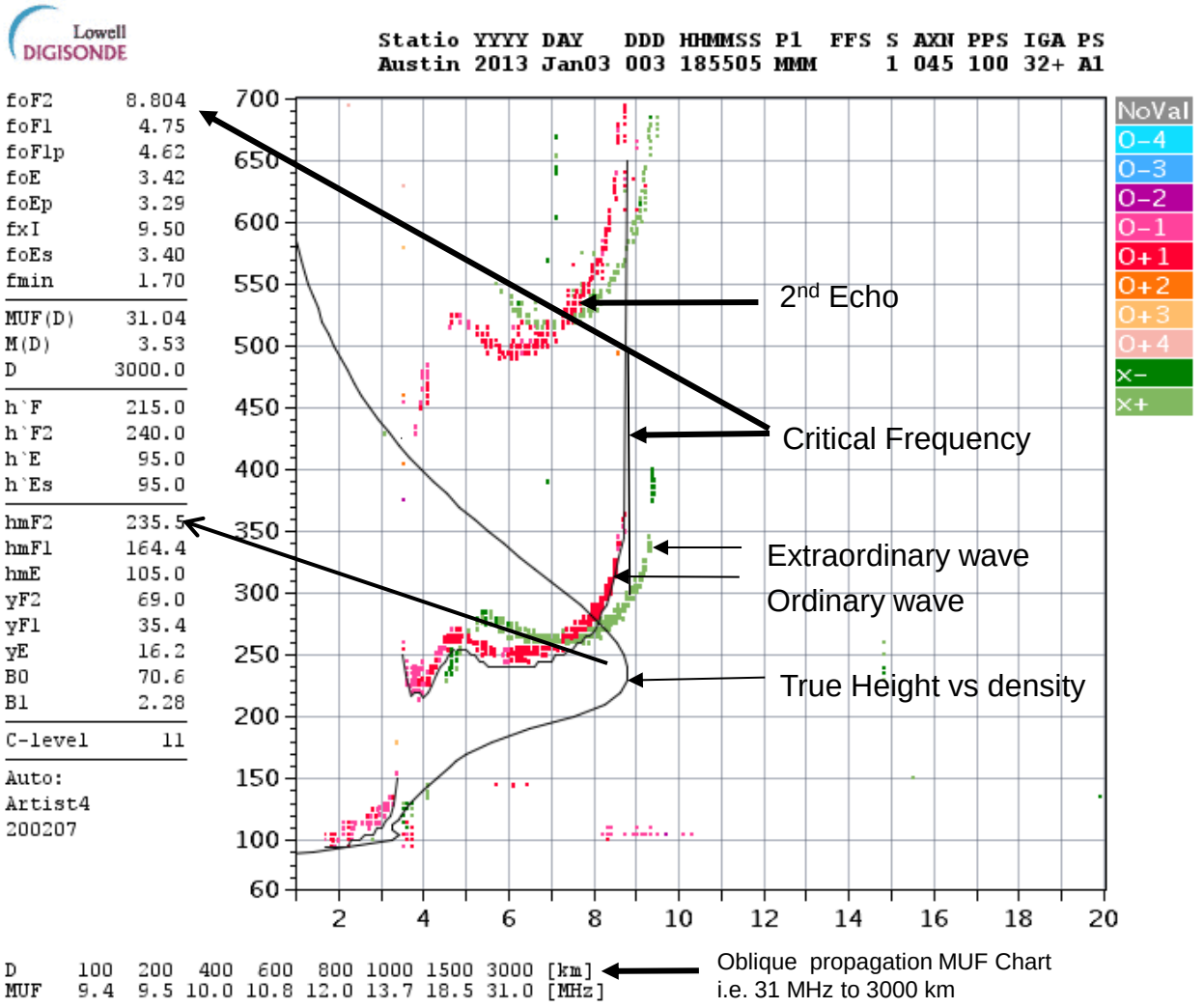
Operating Frequency

- Must Operate at below the local **Critical Frequency (CF) for the F2 Layer.**
- To minimize D-layer absorption during day-light hours, operate as close to CF as possible.
- To operate 24 hours a day, you must have a frequency agile antenna system, i.e. 75m to 30m (3.8 MHz – 10 MHz).
- Propagation modeling programs do not work.
- Use the Austin or Eglin GAMBIT plots to obtain CF.
- If not available, use radio checks with close stations (50-100 miles).

Factors Affecting Critical Frequency

- Critical Frequency increases with increased ionization of F-layers
  - Time of day
  - Time of year (Summer Anomaly)
  - Time of the 11-year sun spot cycle
- Other factors
  - Geomagnetic storms (CME and Coronal Holes)
  - Terrestrial heating (summer anomaly)
- NVIS propagation is very sensitive to the correct selection of the Critical Frequency.

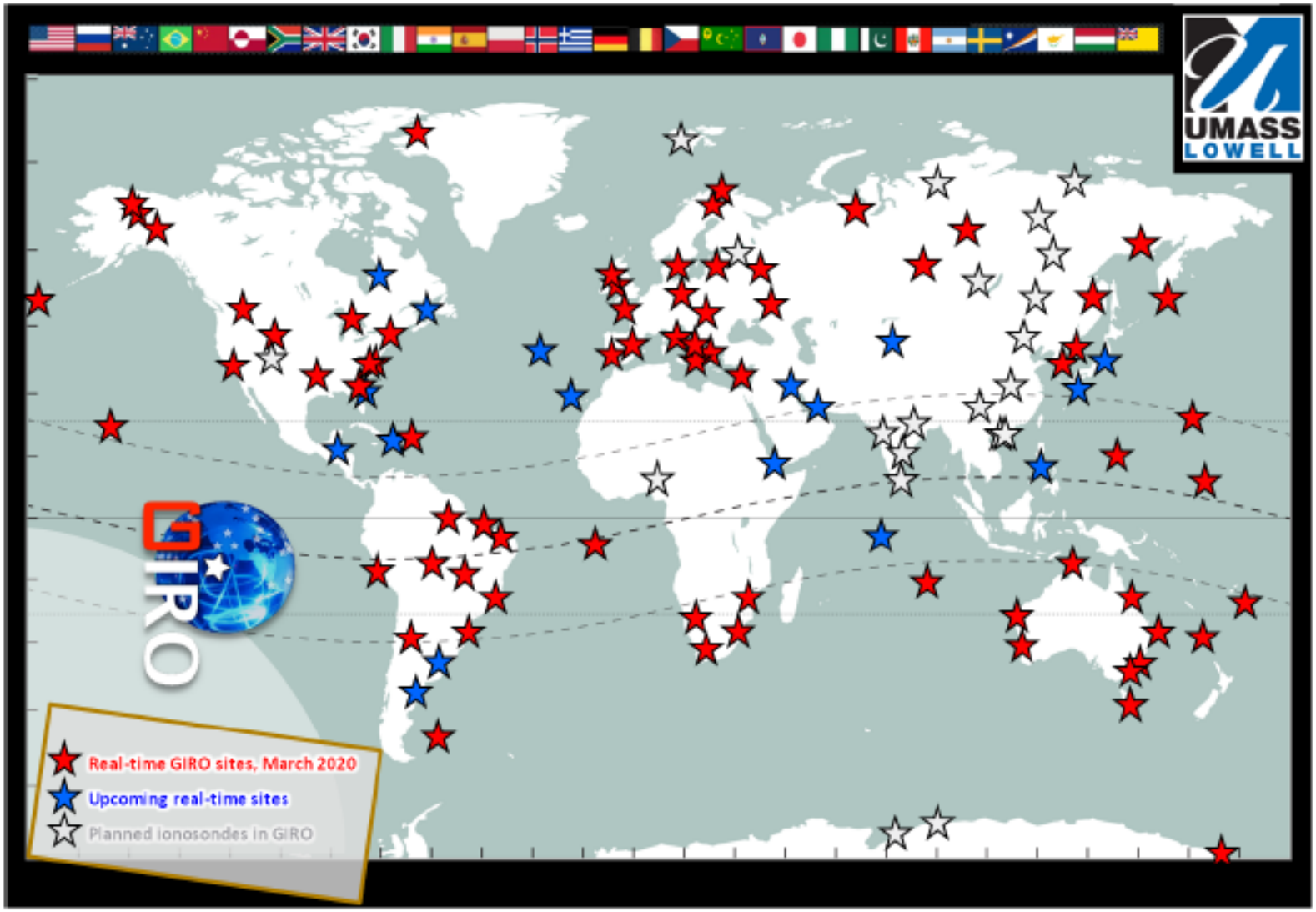
Ionosonde Critical Frequency Measurement



GAMBIT Ionosonde Data Product

- **NOAA is no longer supporting Ionosonde data distribution**.
- The University of Massachusetts, Lowell's GIRO (global Ionosphere Radio Observatory) Data Center provides a number of useful products that can assist HF radio operators in selecting the best HF operating frequency.
- Their GAMBIT (Global Assimilative Model of Bottomside Ionosphere Timeline) product uses IRTAM (IRI Real Time Assimilative Mapping) model to compute foF2 and other useful products every 15 minutes using **real time data** from their world wide Ionosonde data base.
- For more details on IRTAM, See: https://giro.uml.edu/IRTAM/
- I have compared the Eglin Ionosonde (EG931) FTP direct feed Ionograms to GAMBIT plot for EG931 and found very good agreement.

Ionosonde Locations Used By GIRO



GAMBIT

Global Assimilative Model of Bottomside Ionosphere Timeline

GAMBIT SITUATION ROOM

GAMBIT Local Nowcast

Choose a site or type coordinates below:

Location: AU930 AUSTIN

Choose a characteristic:

Characteristic: foF2

Coordinates:

30.40 N (-90..90)

262.30 E (0..360)

☑ Use Current Date and Time

Date mm / dd / yyyy

Submit



Released under CC-BY-NC-SA 4.0, see Rules of the Road for details.

If shared or published, specific data provider must be credited, see Acknowledgement List.

INTERNATIONAL IONOSPHERE · IRTAM · LDI LOWELL DIGISONDE INTERNATIONAL · SSL UML SPACE SCIENCE LAB

Global Ionosphere Radio Observatory (GIRO)

© 1997-2021

Use of GAMBIT For foF2 Plots

- Go to https://giro.uml.edu/rix/gambit-local-nowcast/ .
- The GAMBIT page for any Ionosonde can be seen by entering the following:
  - ***Location*** – A pull down screen allows the user to select a specific Ionosonde by name. Actual user location can be also be entered in the *Coordinates* fields. Note that North Latitude is positive sign. Longitude is in East notation requiring the user to subtract his West longitude from 360. For example -97.8˚ W = 262.2˚ E. For stations with East Longitude, simple enter East longitude value.
  - ***Characteristic*** *fields available:*
    - ***foF2*** **- Critical Frequency, MHz**
    - *NmF2* – Electron density in the F2 layer (not important)
    - ***hmF2*** **– Height of the F2 layer, Km**
    - *B0*, B1 – IRI vertical profiles (not important)

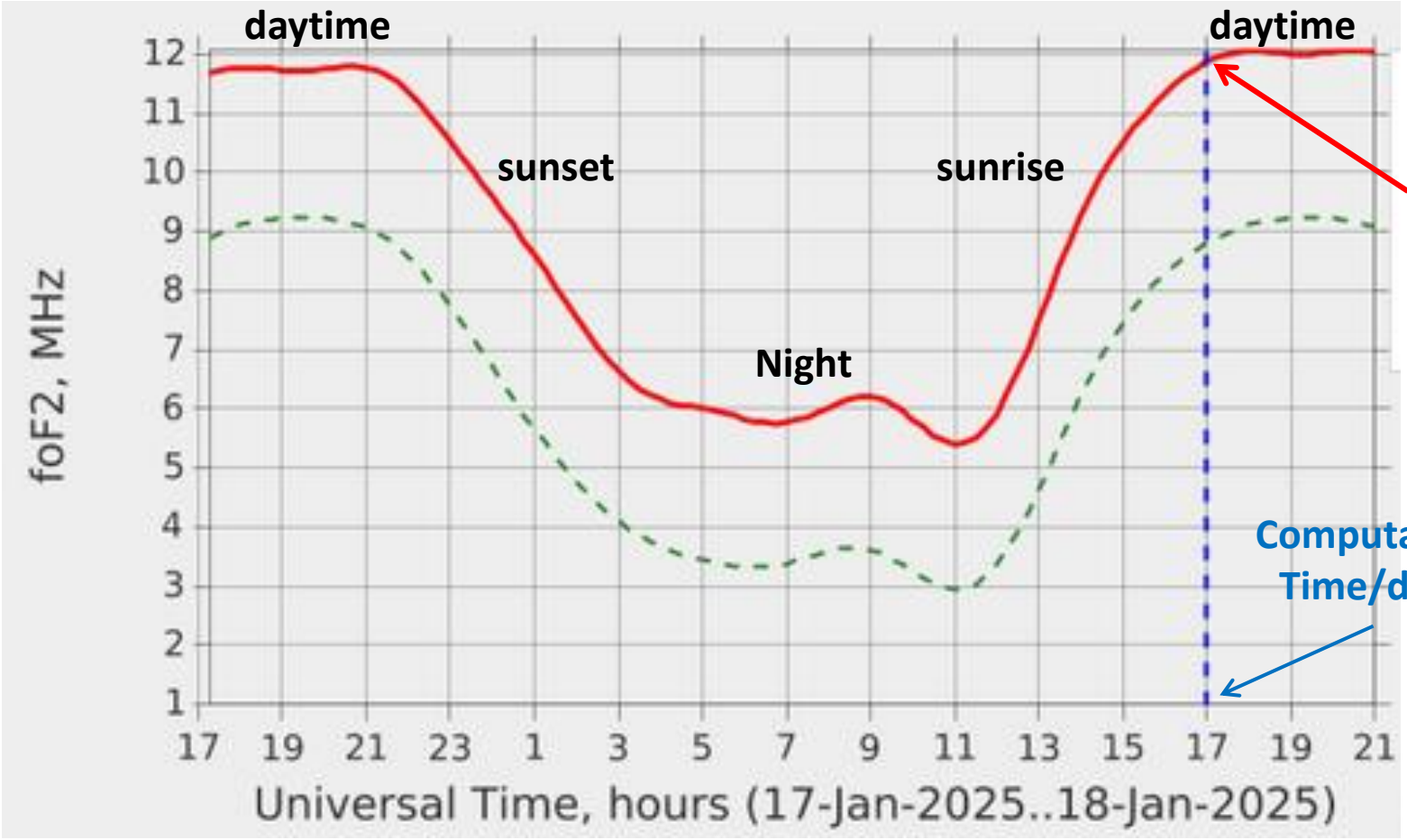
Select *– current date and time (you do not need to enter date and time)*

Click *– Submit*

If the GIRO server is available, the plot on page 18 will be generated. If no result wait and try again later. I would recommend copying this plot for use for the next 24 hours if no severe Ionospheric storms are forecasted. Note that date/time on the horizontal axis is in UTC and can include a change in date.

This plot can be easily copied and printed using *Snipping Tool* or similar graphical capture programs. If you distribute this data please include the source (see rules of the road).

GAMBIT foF2 Trending Chart for Austin Ionosonde



GAMBIT Location on Region 6 Army MARS Solar Weather Page

https://www.region6armymars.org/resources/solarweather.php

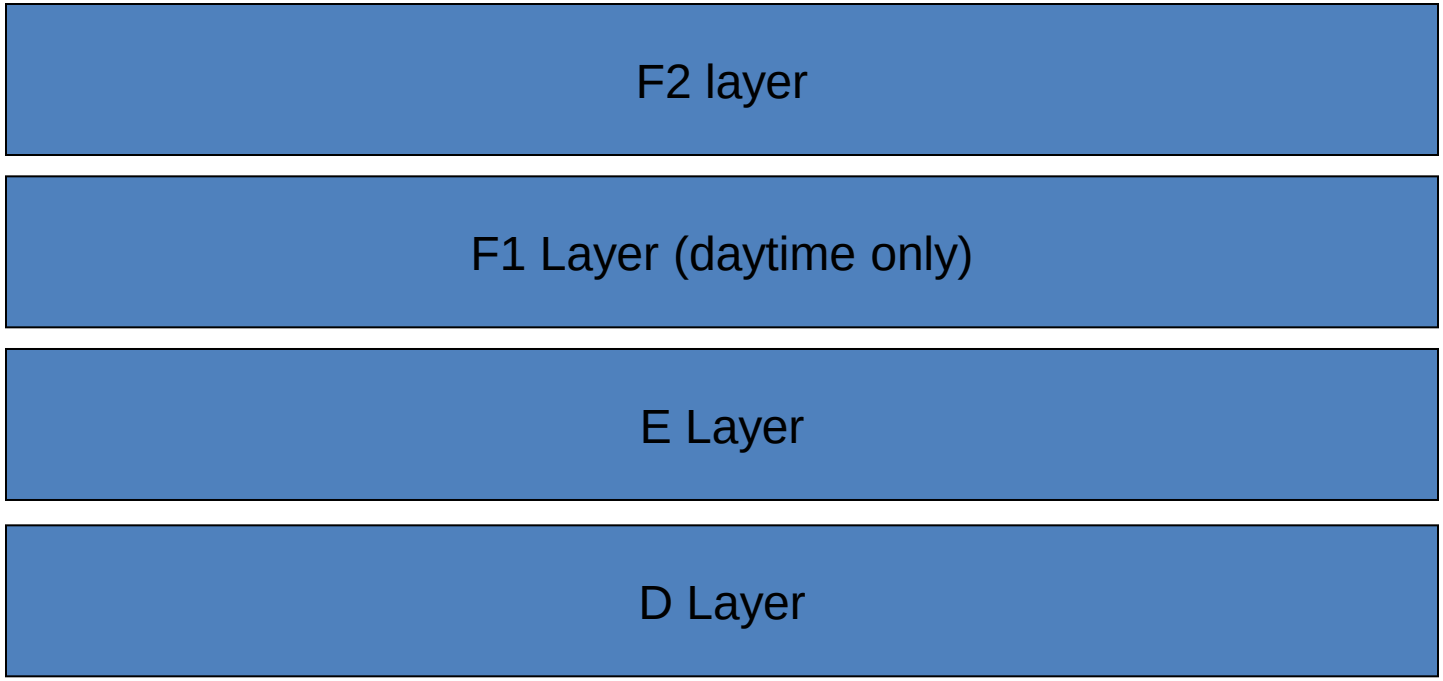
- GAMBIT - Global Assimilative Model of Bottomside Ionosphere Timeline (World-Wide)
  - Boulder
  - Eglin

- NOAA Solar Weather - Solar Weather plots of Kp and X-Ray and other solar emissions.
- Solen Solar Weather - Good general solar forecast from an individual.
- Solar Ham - SolarHam provides real time solar news, as well as consolidated data from various sources.

D-Layer Absorption

- Why not just operate at a frequency much lower than the CF?
- OK for night-time nets but during day-light hours, D-layer absorption will reduce signal levels dramatically at lower frequencies.

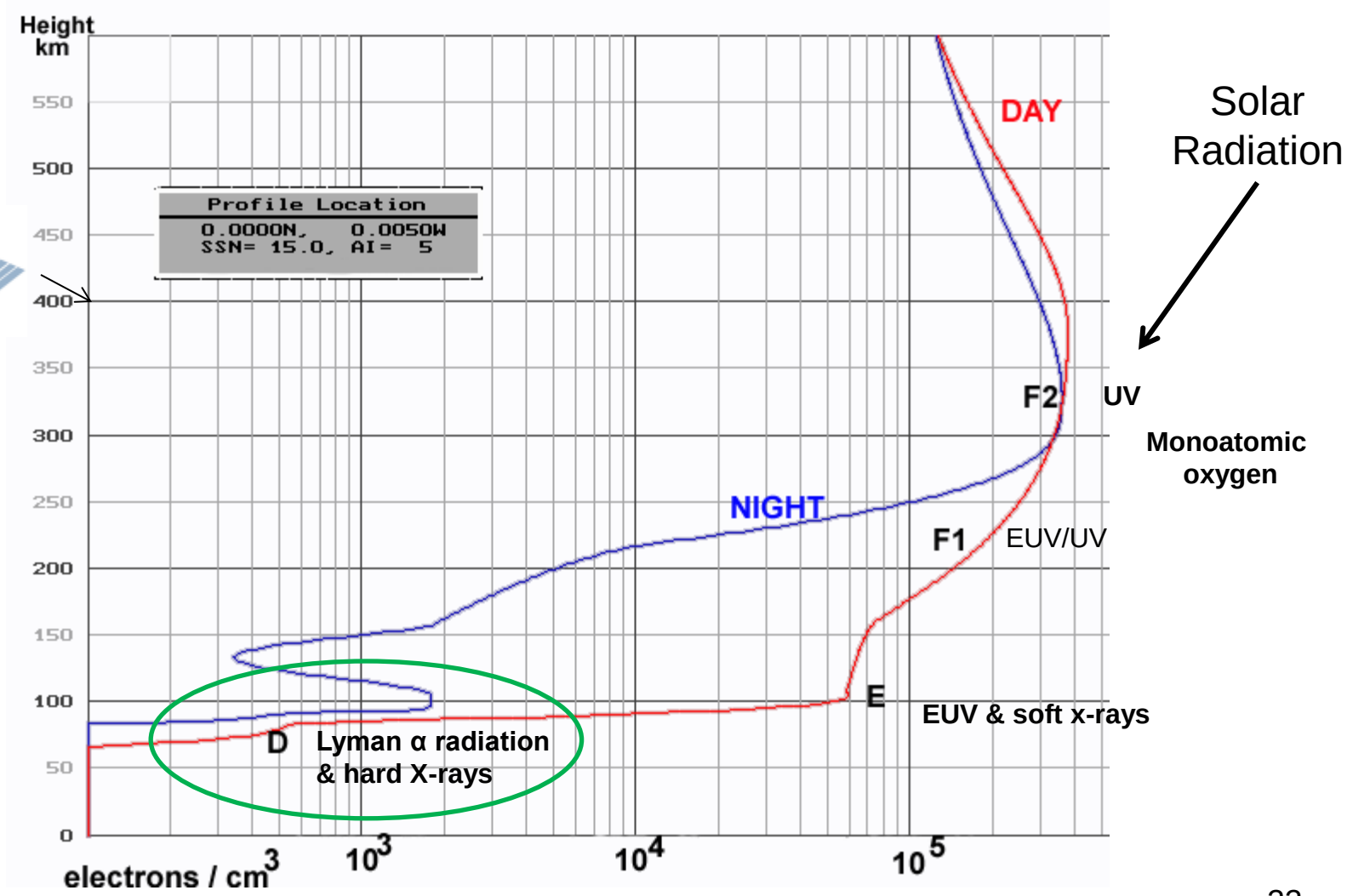
IONOSPHERE



Ionosphere Structure



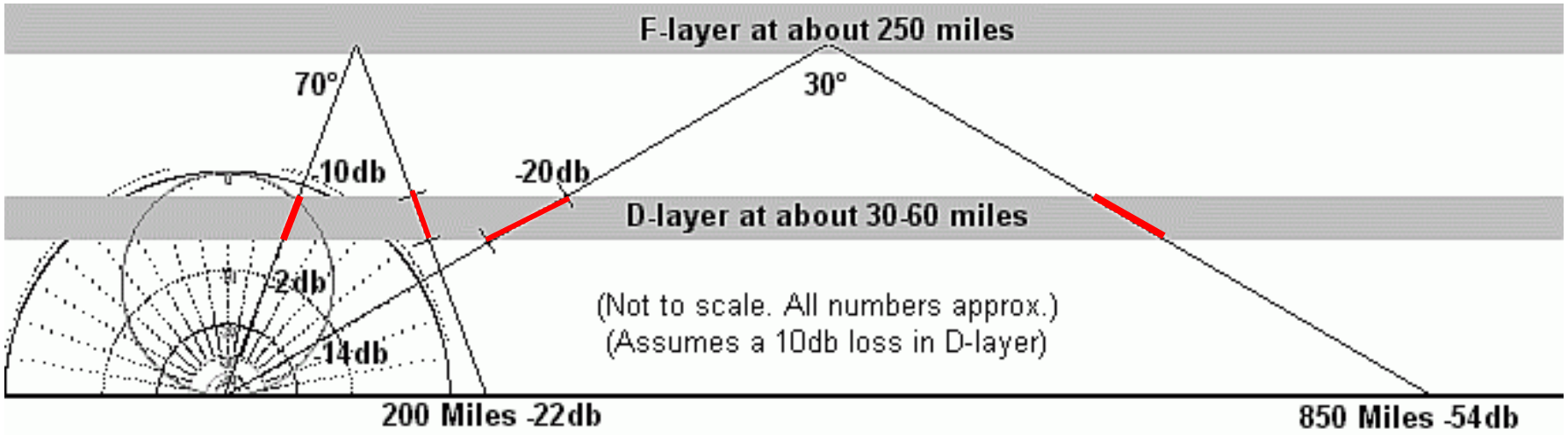
Gravity



Solar Radiation

Lowest Usable Frequency (LUF)

- Controlled by D-Layer absorption
- Day-time effect
- Function of transmit power and mode of operation
- Absorption is a function of $1/f^2$



Frequency Selection Rule

- Day-time: Operate at or just below the Critical Frequency at all times to minimize D-layer absorption.
- Night-time: Operate at or below CF to maintain NVIS propagation as CF drops.
- In both cases, NCS may need to change the net frequency several times during a 30 minute net.

NVIS Antenna Design

- Operating Frequency
- Antenna design
  - Correct directivity pattern
  - Provide usable impedance to RF transmitter

Frequency versus Wavelength

- When designing antennas, the length and height in wavelengths determines the antenna characteristics. It is easier to work in wavelengths and then convert to actual dimensions at the design frequency(s):

  λ (wavelength, ft.) = 983.6/f (frequency, MHz)

- A convenient lookup table follows.

Frequency to Wavelength Chart

Frequency (kHz)		Wavelength (Meters)		Wavelength (Feet)		1/2 Wavelength (Feet)		1/4 Wavelength (Feet)		1/3 Wavelength (Feet)		1/10 Wavelength (Feet)	
2,194.00	2,495.00	136.7	120.2	448.6	394.5	224.3	197.2	112.2	98.6	149.5	131.5	44.9	39.4
2,505.00	2,850.00	119.8	105.3	392.9	345.3	196.5	172.7	98.2	86.3	131.0	115.1	39.3	34.5
3,155.00	3,400.00	95.1	88.2	312.0	289.5	156.0	144.7	78.0	72.4	104.0	96.5	31.2	28.9
4,000.00	4,063.00	75.0	73.8	246.1	242.2	123.0	121.1	61.5	60.6	82.0	80.7	24.6	24.2
4,438.00	4,650.00	67.6	64.5	221.8	211.7	110.9	105.8	55.4	52.9	73.9	70.6	22.2	21.2
4,750.00	4,995.00	63.2	60.1	207.2	197.0	103.6	98.5	51.8	49.3	69.1	65.7	20.7	19.7
5,005.00	5,450.00	59.9	55.0	196.7	180.6	98.3	90.3	49.2	45.1	65.6	60.2	19.7	18.1
5,730.00	5,950.00	52.4	50.4	171.8	165.4	85.9	82.7	42.9	41.4	57.3	55.1	17.2	16.5
6,765.00	7,000.00	44.3	42.9	145.5	140.6	72.7	70.3	36.4	35.2	48.5	46.9	14.5	14.1
7,300.00	8,195.00	41.1	36.6	134.8	120.1	67.4	60.1	33.7	30.0	44.9	40.0	13.5	12.0
9,040.00	9,500.00	33.2	31.6	108.9	103.6	54.4	51.8	27.2	25.9				
9,900.00	9,995.00	30.3	30.0	99.4	98.5	49.7	49.2	24.9	24.6				
10,150.00	11,175.00	29.6	26.8	97.0	88.1	48.5	44.0	24.2	22.0				
11,400.00	11,650.00	26.3	25.8	86.3	84.5	43.2	42.2	21.6	21.1				
12,050.00	12,230.00	24.9	24.5	81.7	80.5	40.8	40.2	20.4	20.1				
13,410.00	13,600.00	22.4	22.1	73.4	72.4	36.7	36.2	18.3	18.1				
13,800.00	14,000.00	21.7	21.4	71.3	70.3	35.7	35.2	17.8	17.6				
14,350.00	14,990.00	20.9	20.0	68.6	65.7	34.3	32.8	17.1	16.4				
15,600.00	16,360.00	19.2	18.3	63.1	60.2	31.5	30.1	15.8	15.0				
17,410.00	17,550.00	17.2	17.1	56.5	56.1	28.3	28.0	14.1	14.0				
18,030.00	18,068.00	16.6	16.6	54.6	54.5	27.3	27.2	13.6	13.6				
18,168.00	18,780.00	16.5	16.0	54.2	52.4	27.1	26.2	13.5	13.1				
18,900.00	19,660.00	15.9	15.3	52.1	50.1	26.0	25.0	13.0	12.5				
19,800.00	19,990.00	15.2	15.0	49.7	49.2	24.9	24.6	12.4	12.3				
20,010.00	21,000.00	15.0	14.3	49.2	46.9	24.6	23.4	12.3	11.7				
21,855.00	23,200.00	13.7	12.9	45.0	42.4	22.5	21.2	11.3	10.6				
23,350.00	24,890.00	12.8	12.1	42.2	39.5	21.1	19.8	10.5	9.9				
25,330.00	25,550.00	11.8	11.7	38.9	38.5	19.4	19.3	9.7	9.6				
26,480.00	28,000.00	11.3	10.7	37.2	35.2	18.6	17.6	9.3	8.8				
29,800.00	30,000.00	10.1	10.0	33.0	32.8	16.5	16.4	8.3	8.2				

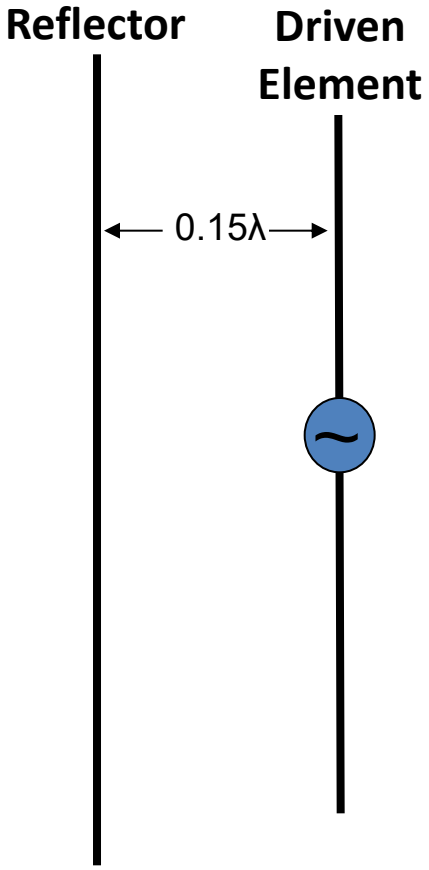
Minimum NVIS Height

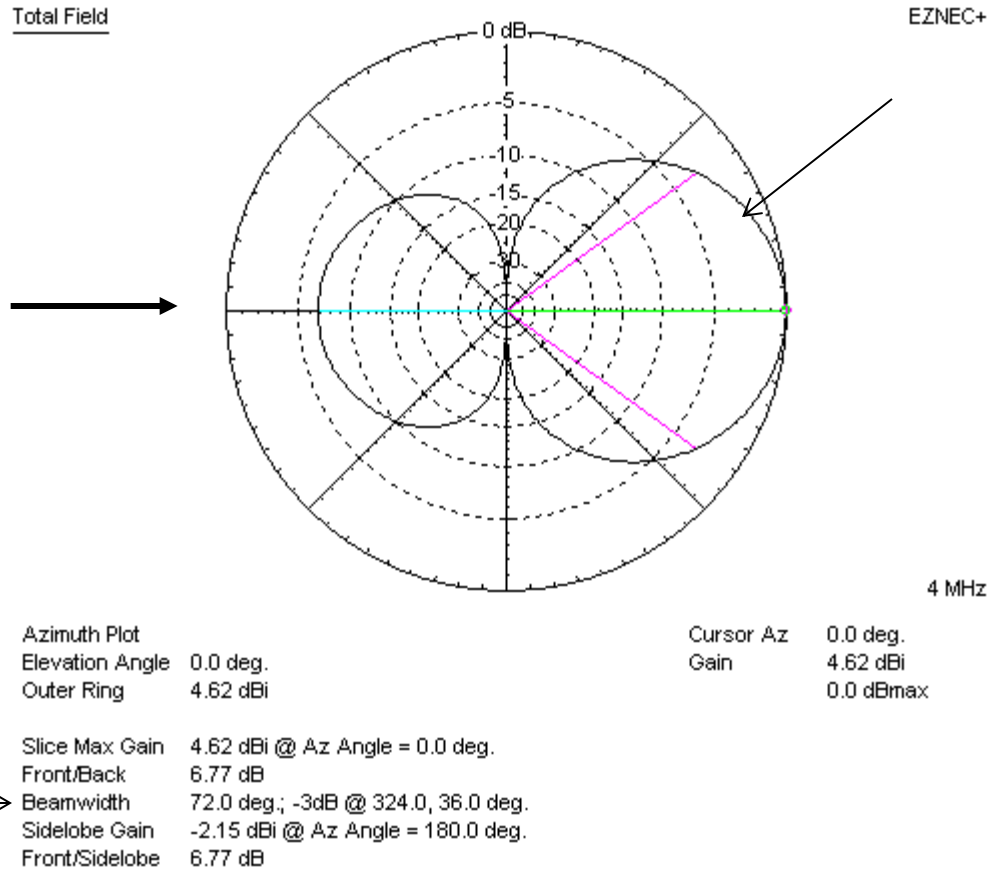
Maximum NVIS Height

Long-Range Height

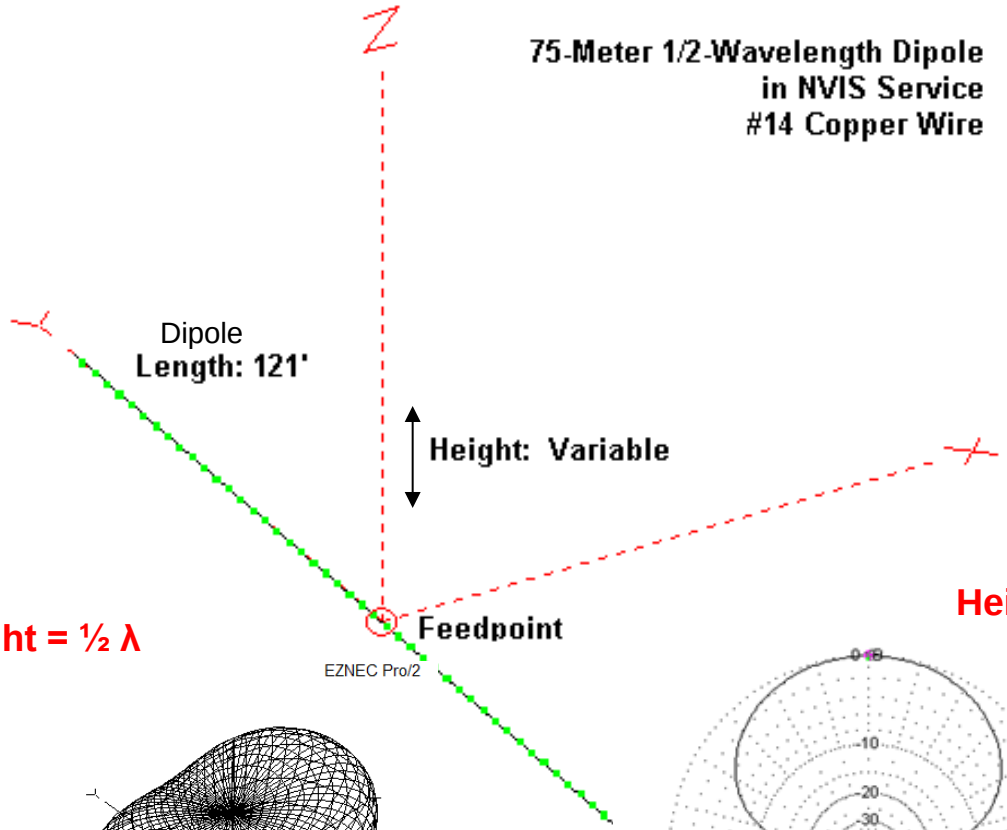
Creating A NVIS Pattern (Theory)

(Free Space)

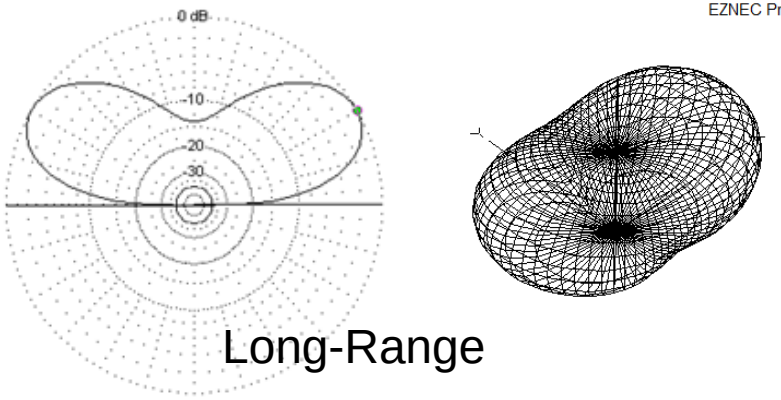




Dipole Directivity versus Height
(1/2 Wavelength Dipole)

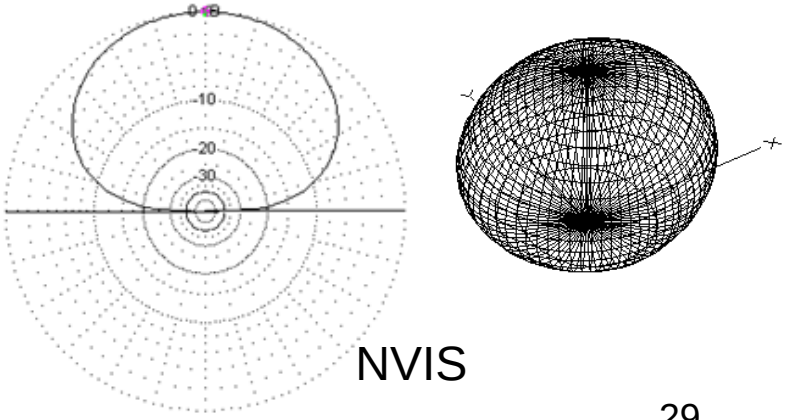


Height = ½ λ



Long-Range

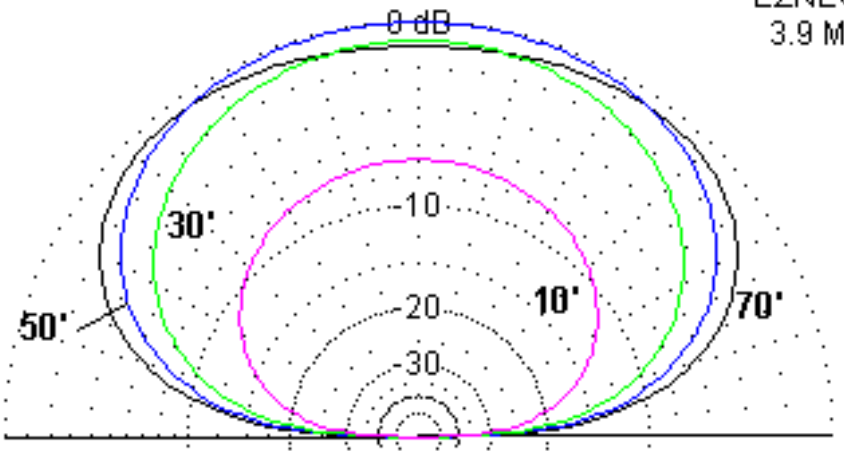
Height = 0.15 λ



NVIS

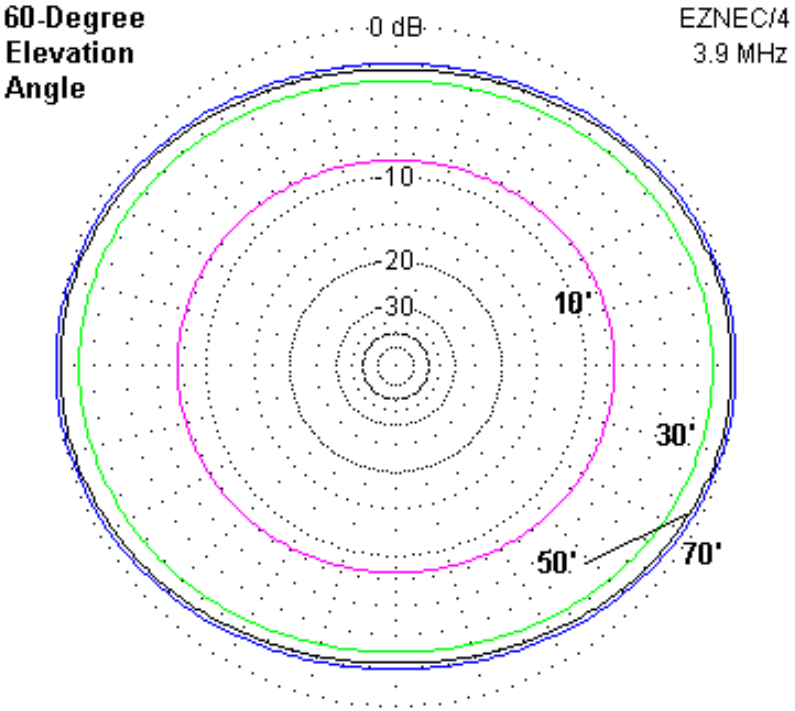
NVIS Dipole Patterns Versus Height – 3.9 MHz

How Low can you go?



Elevation Patterns of a 75-Meter Dipole for NVIS Service at 10, 30, 50, and 70 Feet Above Average Soil

50 ft. = 0.2λ
30 ft. = 0.12λ



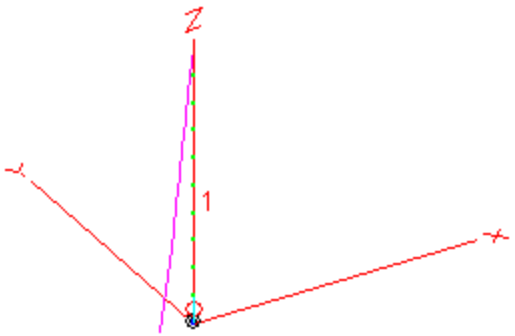
Azimuth Patterns of a 75-Meter Dipole for NVIS Service at 10, 30, 50, and 70 Feet Above Average Soil

Basic NVIS Antenna Rules

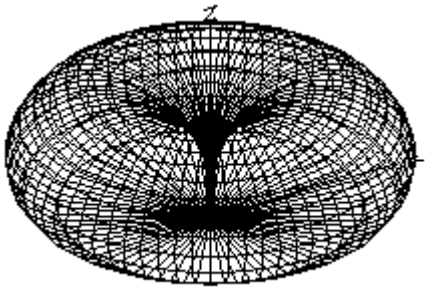
- Excellent antenna is a resonant ½ λ horizontal dipole at 0.15 λ height.
- Good performance from 0.1λ to 0.25λ
- Inverted-V is some 2-3 dB worst than dipole, but has more uniform horizontal pattern and is easier to implement!
- Vertical antennas do not generate a NVIS pattern.

VERTICAL ANTENNAS

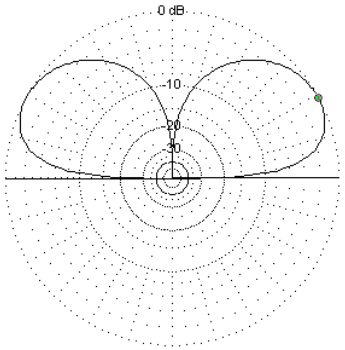
EZNEC



Height = 1.8 m (6 ft)

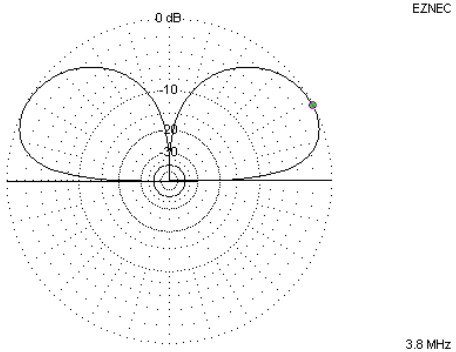


7 MHz



Elevation Plot		Cursor Elev	29.0 deg.
Azimuth Angle	0.0 deg.	Gain	0.29 dBi
Outer Ring	0.29dBi		0.0 dBmax
Slice Max Gain	0.29 dBi @ Elev Angle = 29.0 deg.		
Beamwidth	47.6 deg.; -3dB @ 9.8, 57.4 deg.		
Sidelobe Gain	0.29 dBi @ Elev Angle = 151.0 deg.		
Front/Sidelobe	0.0 dB		

3.8 MHz

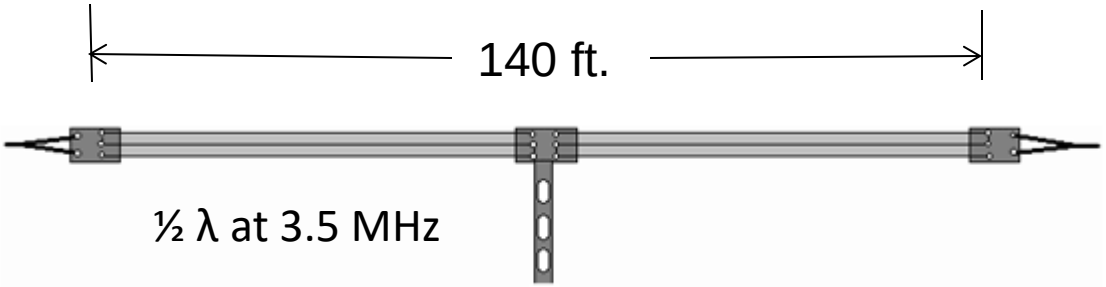


Elevation Plot		Cursor Elev	28.0 deg.
Azimuth Angle	0.0 deg.	Gain	0.92 dBi
Outer Ring	0.92dBi		0.0 dBmax
Slice Max Gain	0.92 dBi @ Elev Angle = 28.0 deg.		
Beamwidth	47.7 deg.; -3dB @ 8.9, 56.6 deg.		
Sidelobe Gain	0.92 dBi @ Elev Angle = 152.0 deg.		
Front/Sidelobe	0.0 dB		

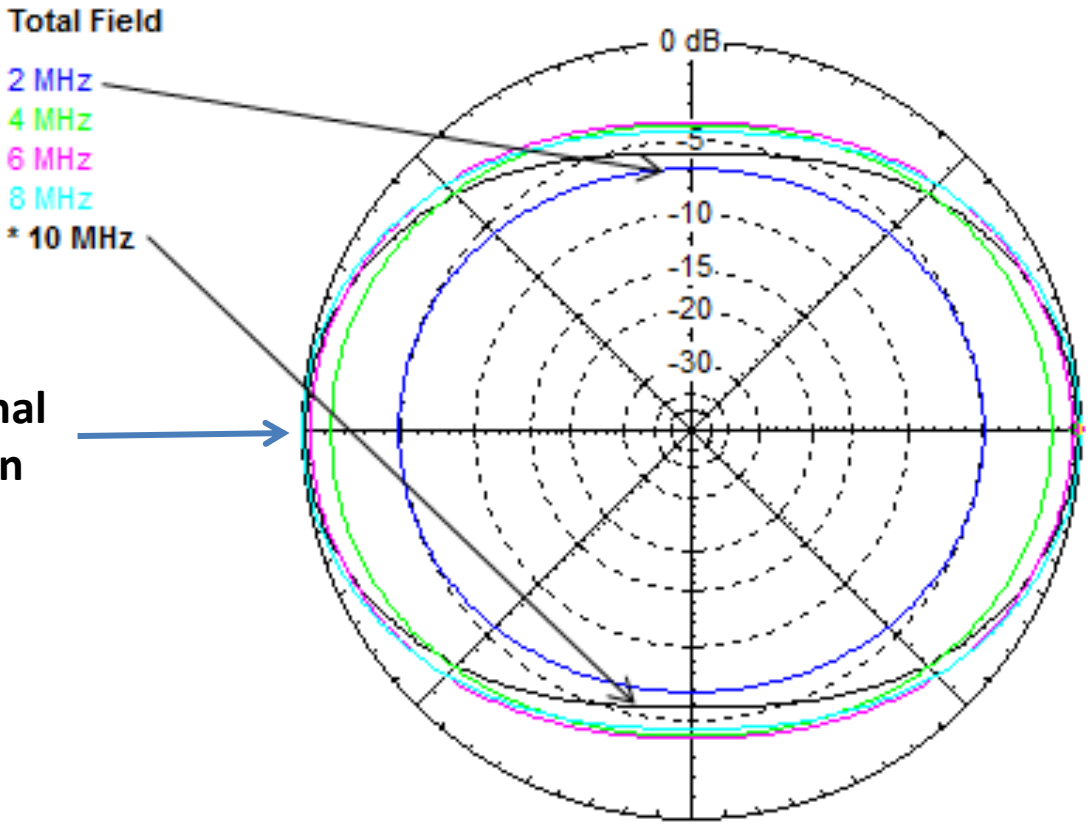
Broadband NVIS Antenna Design

- To operate just below Critical Frequency your antenna needs to operate from 3.8 MHz to 10 MHz amateur bands (75m – 30m).
- When the antenna length exceeds ½ λ, the horizontal (azimuth) NVIS pattern departs from omni-directional response and becomes increasingly more directional even when the correct antenna height is maintained.

Single Broadband Dipole – Cobra Senior



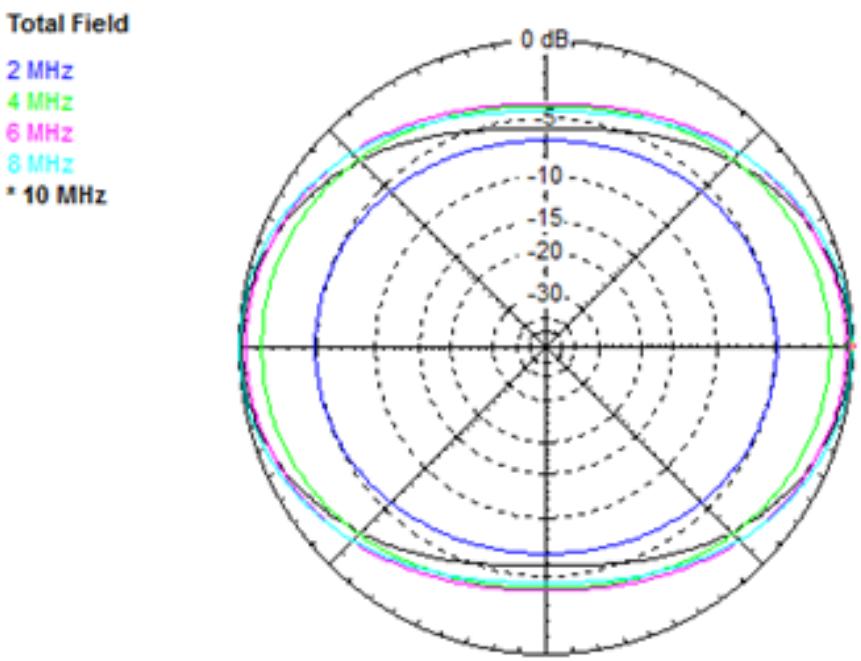
40 ft.



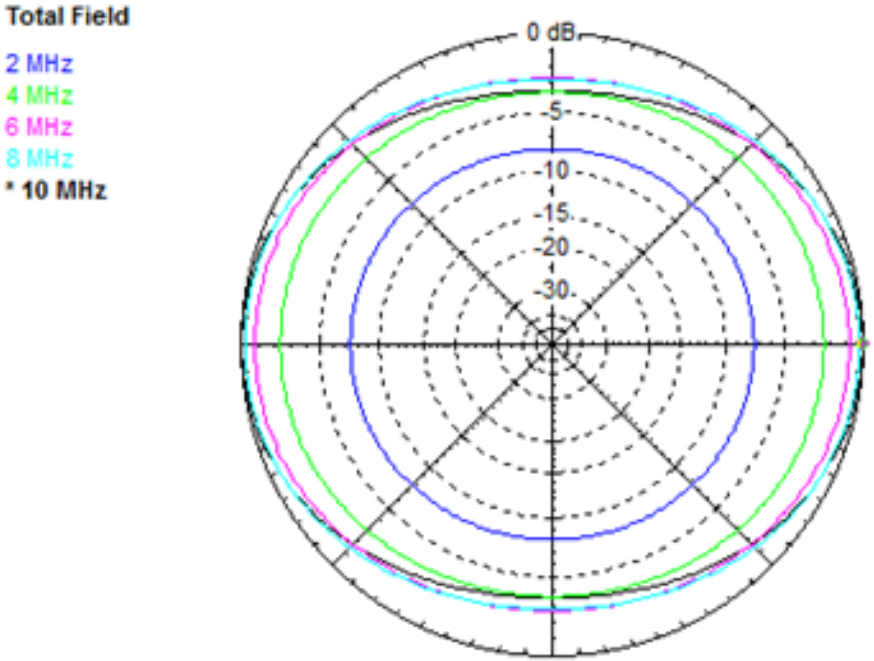
Azimuthal Pattern

Inverted-V Solution

(Cobra Senior - 45° Inverted-V)



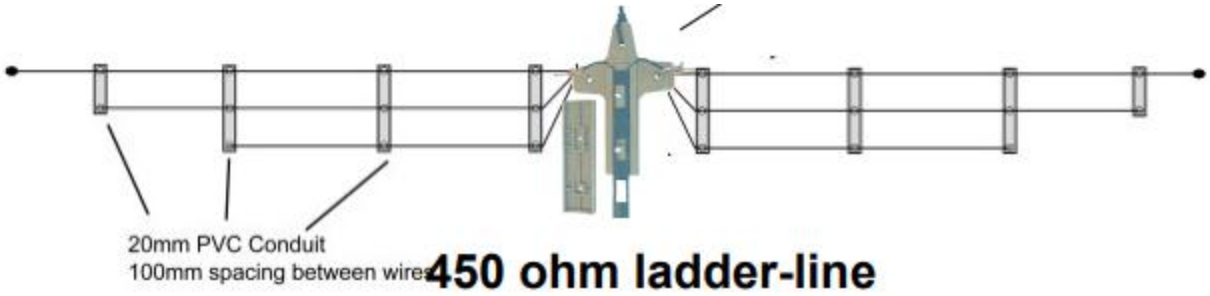
Horizontal Dipole



45° Inverted-V

Other Dipole Directivity Solutions

- Fan Dipole – Shorter legs dominate radiation pattern as frequency increases. Requires careful design including modeling.



- Traps – Shortens antenna length as frequency increases. Can be lossy and difficult to tune.

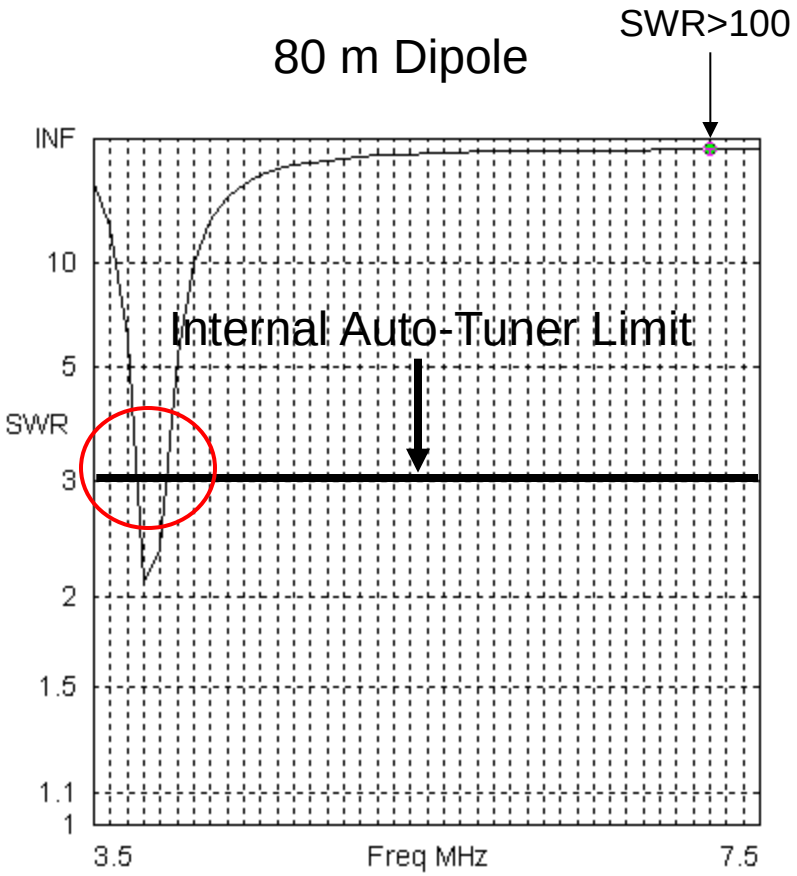


NVIS PARAMETERS

- Antenna design
  - Correct directivity pattern
  - Provide usable impedance to RF transmitter (i.e.,Low VSWR)
    - All modern HF transceivers will start “folding-back” the transmit power when VSWR exceed 1.5:1 and will typically shut off power at 3:1.
    - Some transceivers have built-in tuners but they are limited to 3:1 VSWR correction.

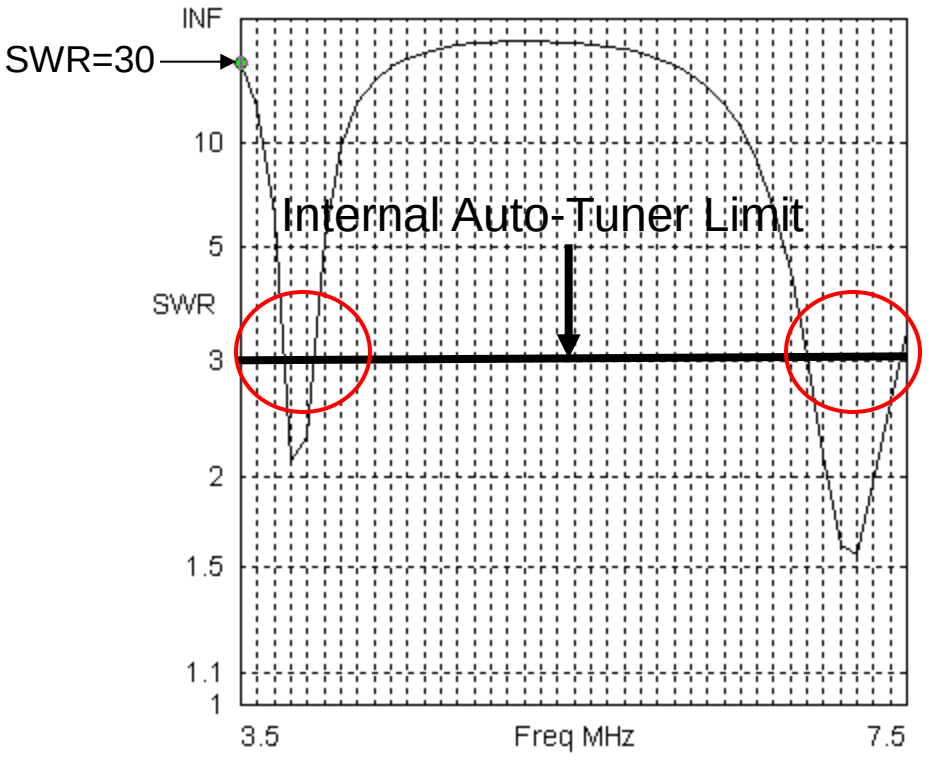
Internal Auto-Tuner Limitations

80 m Dipole



Freq	7.2 MHz	Source #	1
SWR	> 100	Z0	50 ohms
Z	4589 + j 2444 ohms		
Refl Coeff	0.9832 at 0.52 deg.		

80/40 m Cross-Dipoles



Freq	3.5 MHz	Source #	1
SWR	30.3	Z0	50 ohms
Z	15.27 - j 143 ohms		
Refl Coeff	0.9362 at -38.18 deg.		

Antenna Tuning Options

- Modern Amateur Transceivers with auto-tuners require VSWR < 3:1
- Options for matching:
  - “Magic” antennas (dummy loads!)
  - Exact ½ wavelength antennas for each frequency (50 Ω)
  - Wide-Band Terminated Folded Dipole
  - Automatic Antenna Tuner at or near antenna
  - Manual or automatic tuner at rig

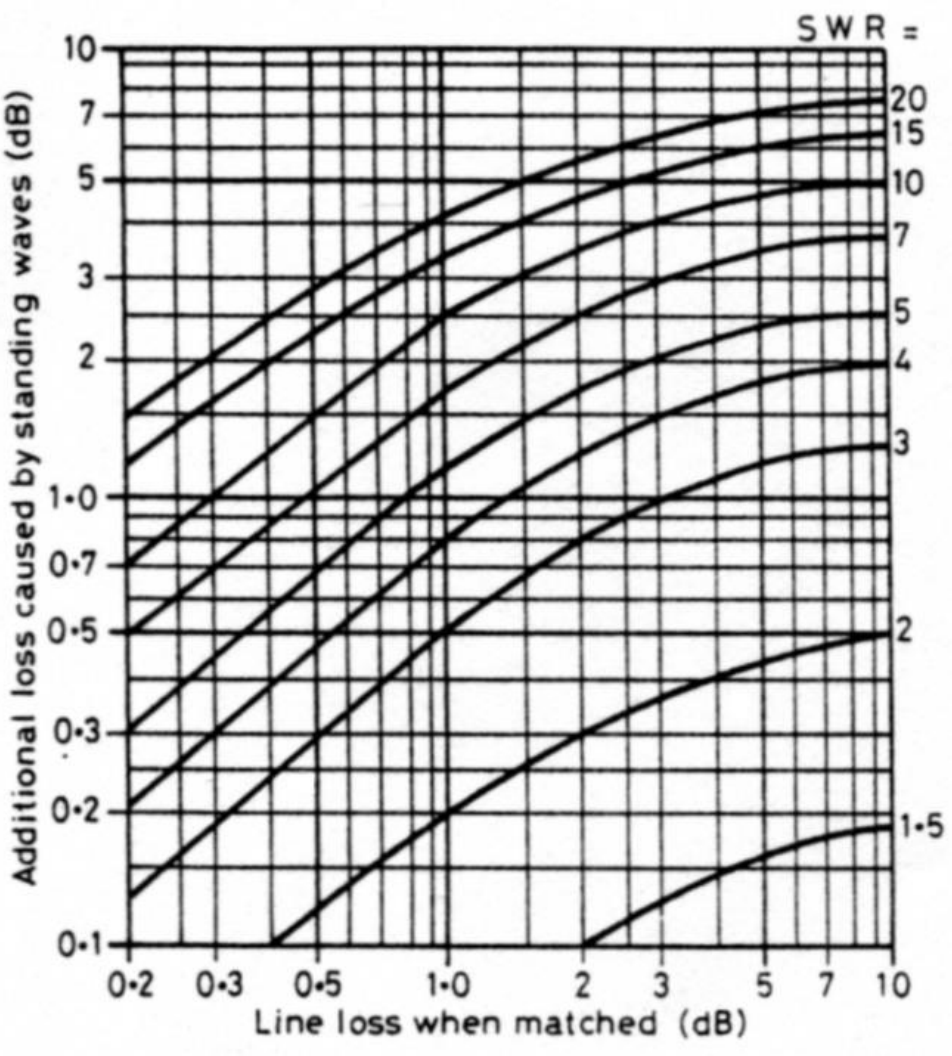
When is VSWR a Concern?

- When using Coaxial Cable at high VSWR.
- Air has no dielectric loss at HF frequencies. This is why ladder is very low loss even at very high VSWR's.

Coaxial Cable Loss

- Additional coaxial cable lose is caused by dielectric losses in the center insulator when cable is operated at high VSWR's.
- Example – If normal line loss is 1 dB and SWR is 20:1, the additional loss is 4 dB for a total loss of 5 dB. 100 watts in = 32 watts out!

- Ladder Line has an air dielectric so has very low losses even at high SWR.



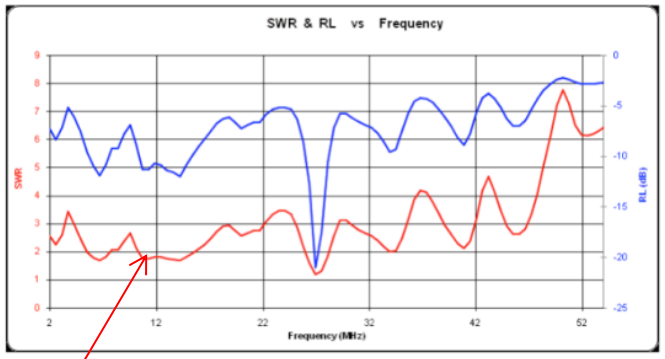
Low VSWR Antennas

- Terminated or loosely coupled Antennas
- One-half wave-length Dipoles
- Special dimensioned wire antennas (for example - G5RV)
- Commercial antennas

Poor Antennas
at NVIS Frequencies (2-6 MHz)

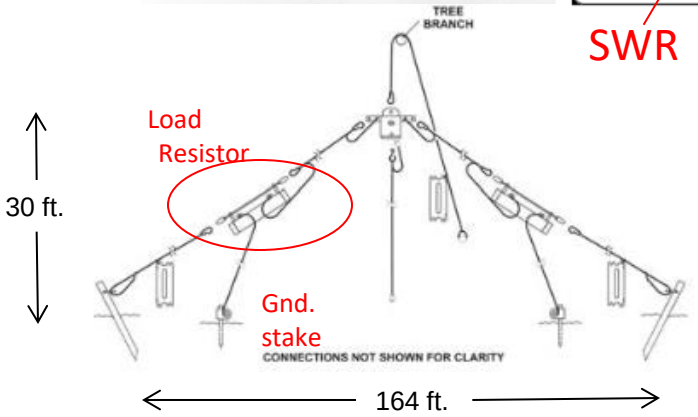
Chameleon EMCOMMII
3.5 – 30 MHz
<2.5:1 SWR
$149.00

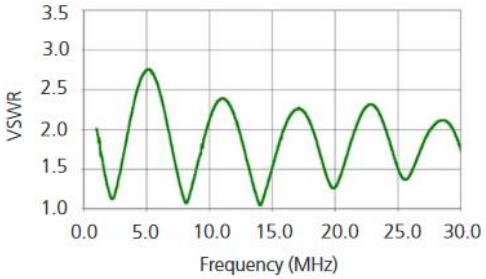




SWR

Harris RF-1944
1.6 – 30 MHz
<2.5:1 SWR
-15 dBi at 2 MHz
(20W in – 0.1 W out)
-2 dBi at 30 MHz
$2530.00





Any long-wire, vertical, or dipole claiming VSWR of <3:1 over the complete HF band has a very high loss matching network and will have a low **transmit efficiency**. Receiving requires only gain so will appear to work as a receiving antenna.

Terminated Folded Dipole (TFD) (B&W 90)

From: http://on5au.be/content/a10/wire/wbfd.html

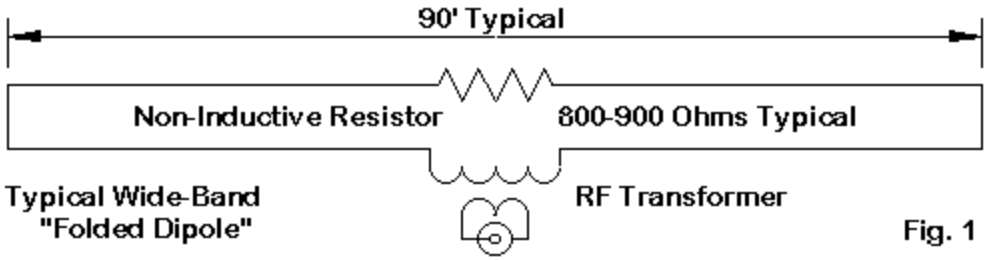


Fig. 1

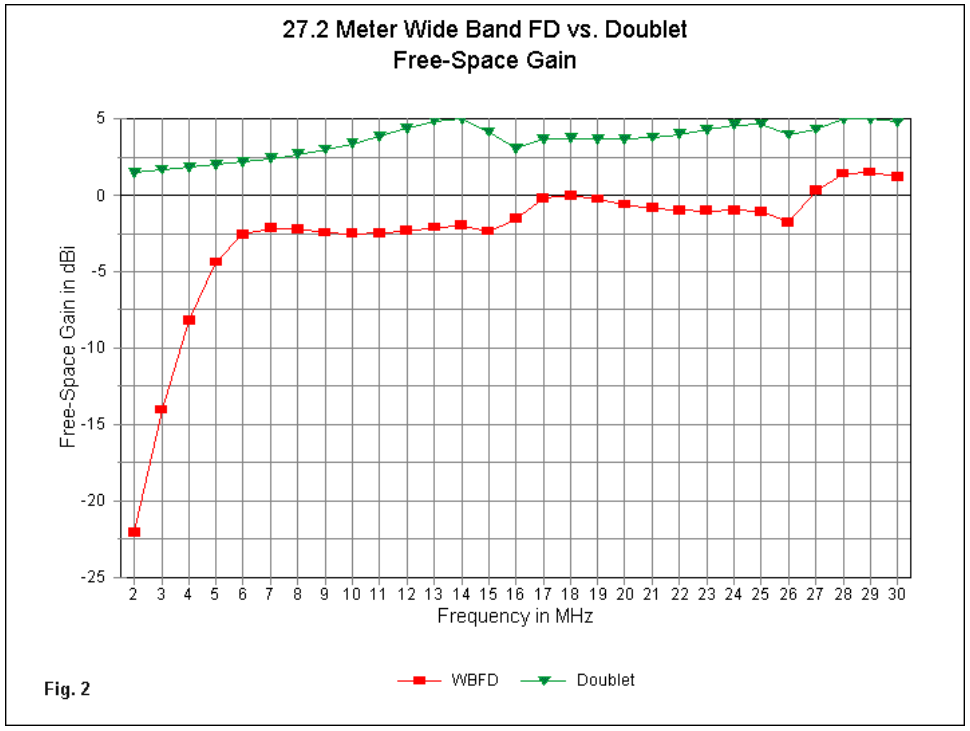


Fig. 2

Rules:

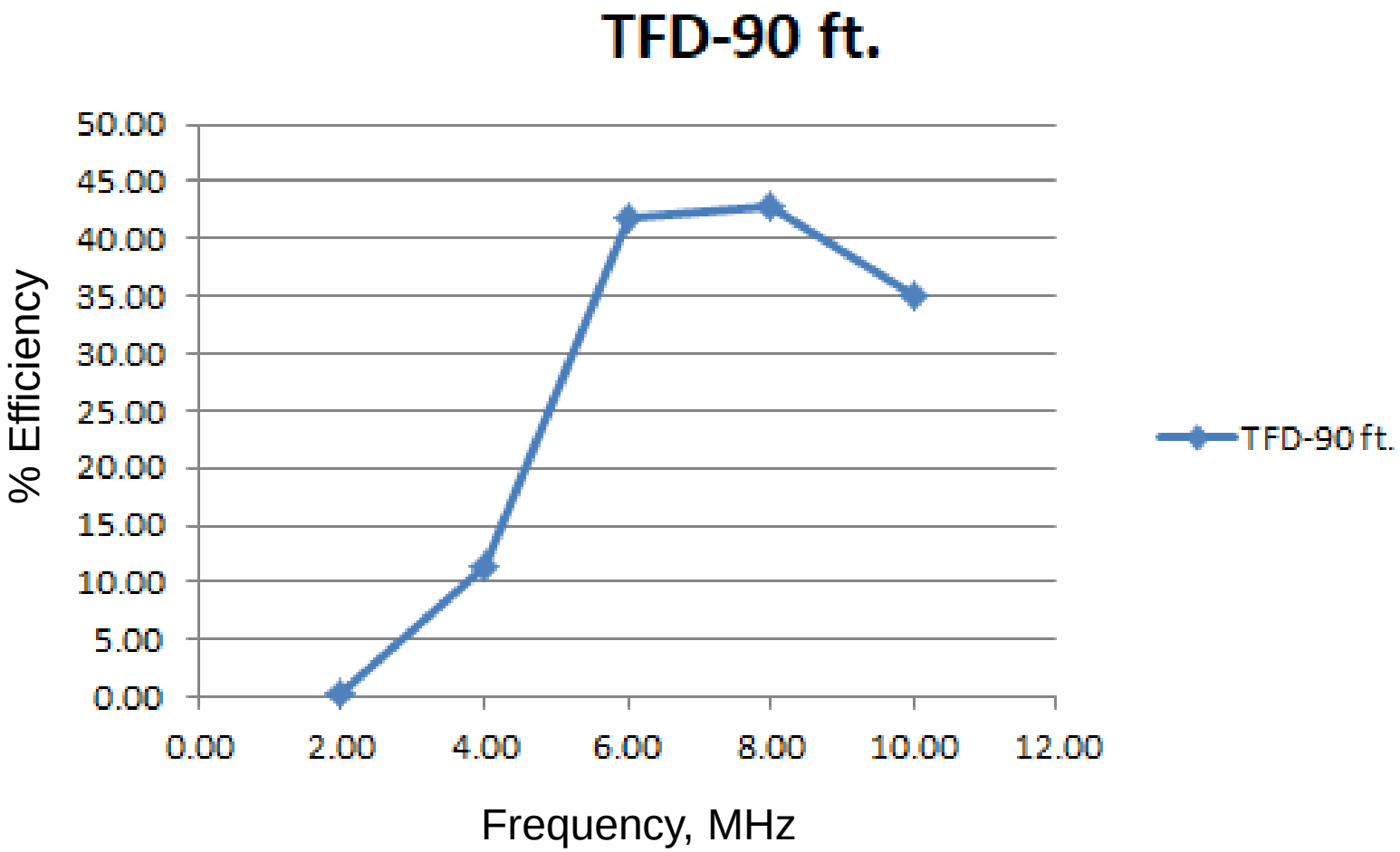
1. Length – ½ wavelength at lowest operating frequency
   B&W 90 > 5 MHz
   B&W 180 > 3 MHz
2. Accept losing ¾ of your transmit power, i.e., 100 watts in, 25 watts out!
3. Solution for ALE

Terminated Folded Dipoles

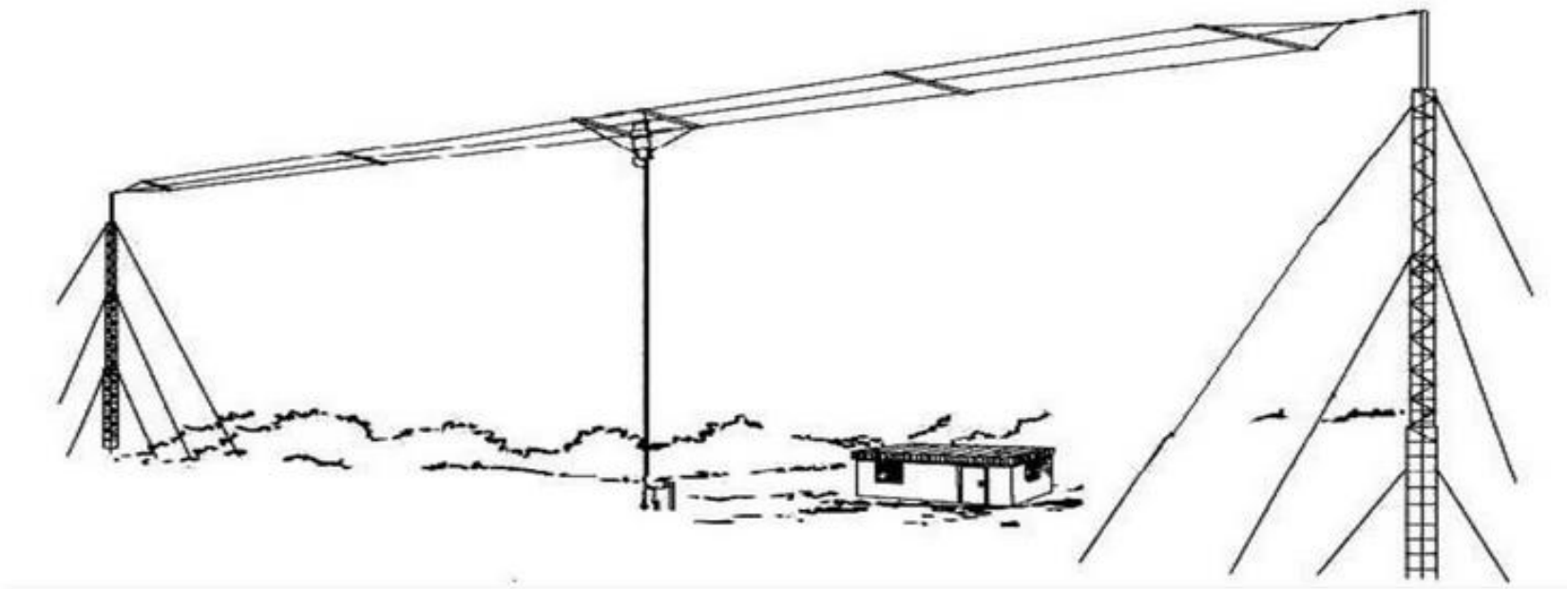


B&W 90

NVIS Antenna Efficiency



Terminated Folded Dipoles

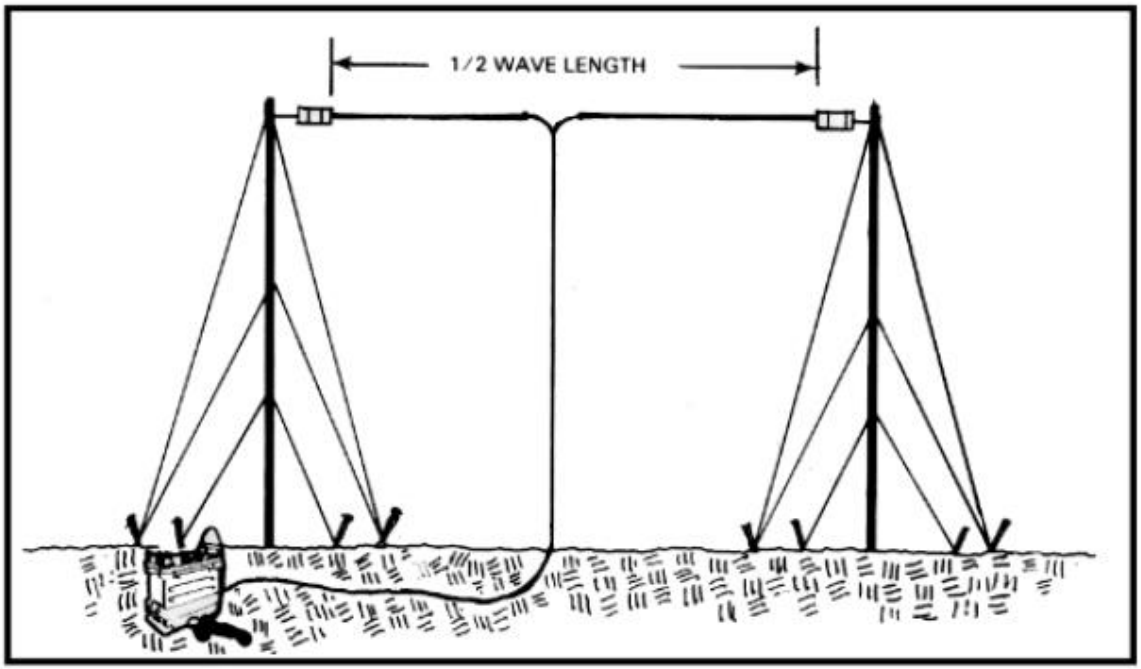


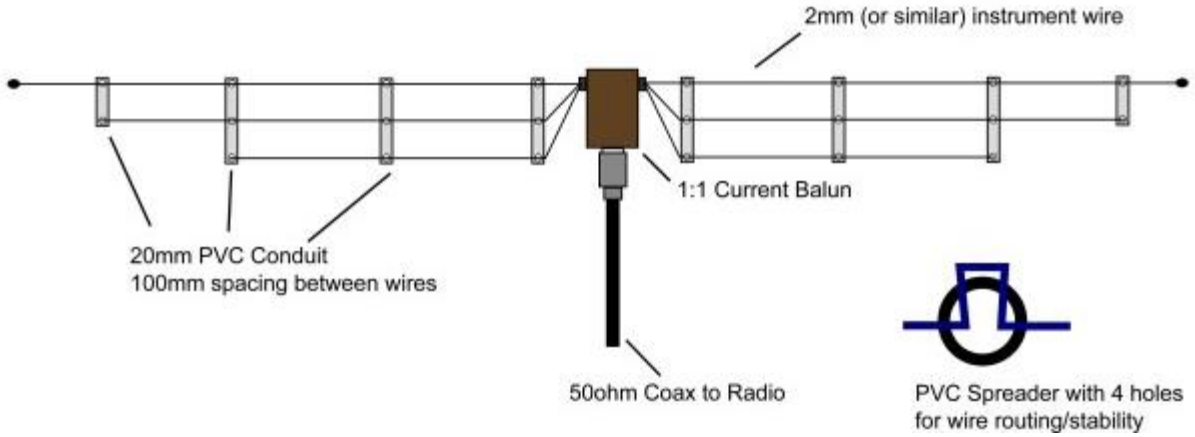
Bushcomm BBA Series

TFD Analyses

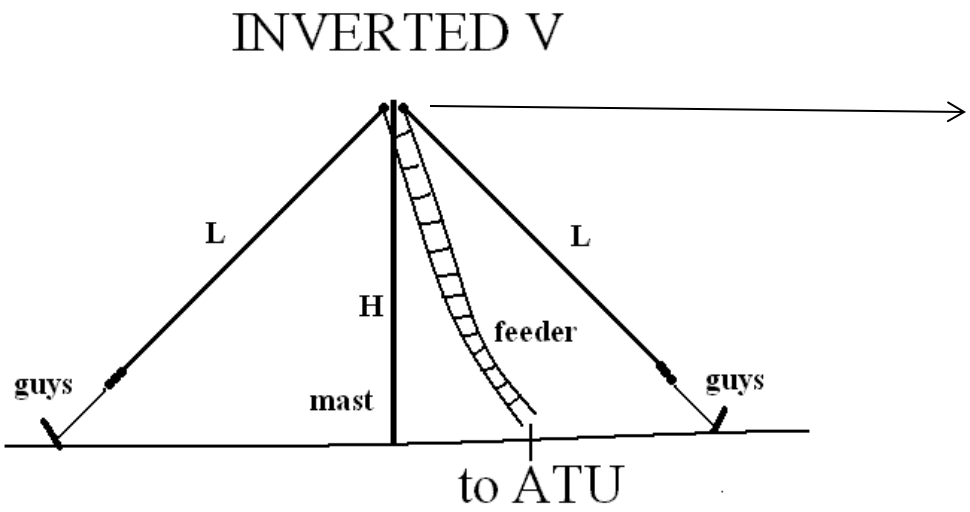
- Have EZNEC Pro files for the following:
  - B&W 90
  - B&W 180
  - Bushcomm BBA 27m (3-wire)
  - Bushcomm BBA 54m (3-wire)
- Only the B&W 180 and BBA 54m are applicable for low NVIS frequencies (3 – 5 MHz). But their length causes severe azimuthal asymmetry at higher frequencies.
- Their low transmit efficiencies also require a HF amplifier to compete with typical non-resonant dipole antennas.
- Only advantage is when a station must rapidly frequency scan, i.e. ALE (Automatic Link Establishment).

Resonant Dipole(s)





Best Option - Non-Resonant Dipole Driven by Tuner





WA1FFL Ladder-Loc

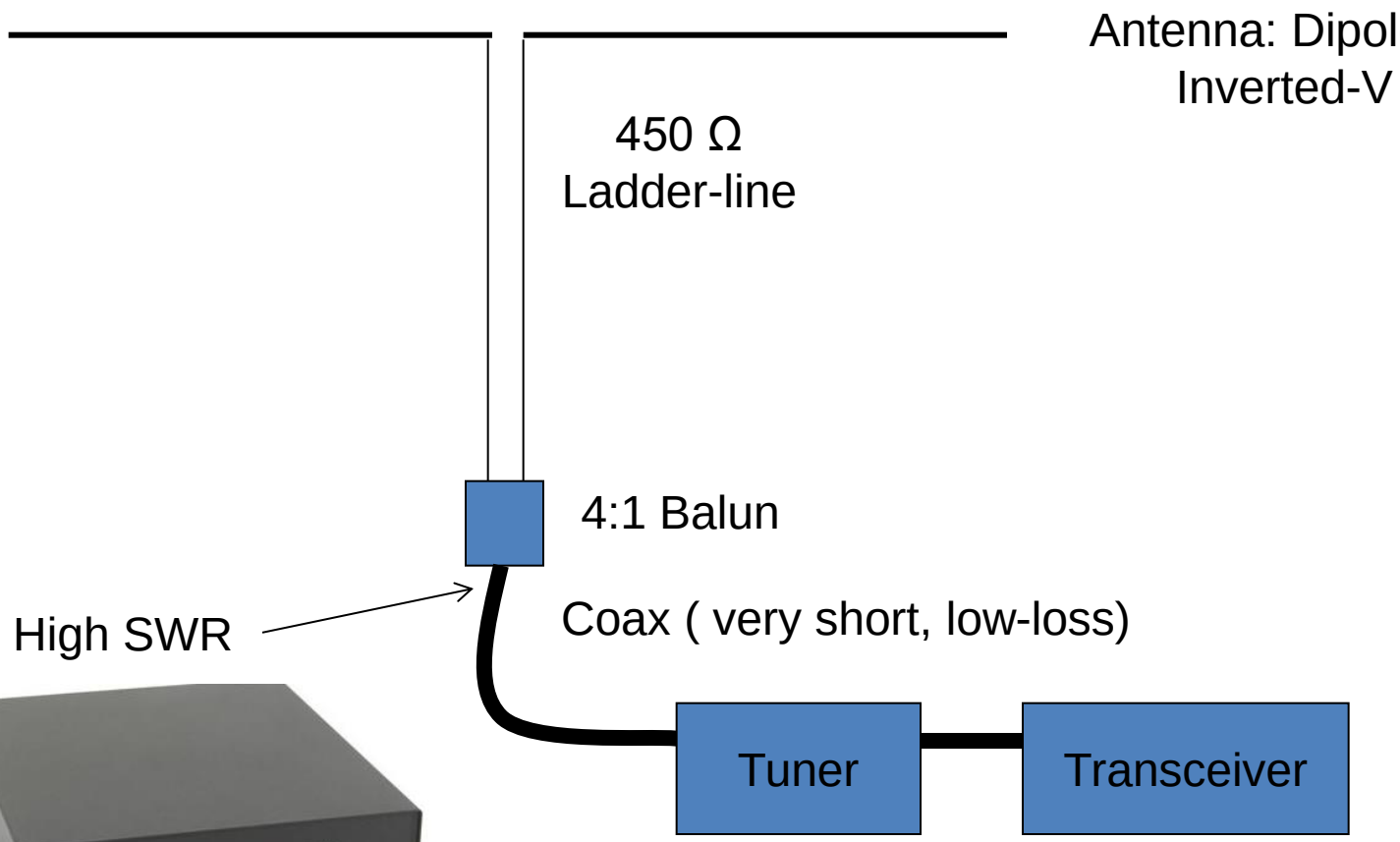
L = 65 ft. – 75 ft.
H = 30 – 50 ft.
Angle = As large as possible

Use 4:1 balun between ladder line and coaxial cable for rig Located tuners.

External Antenna Tuners

- Tuner located at rig
- Tuner located at base of antenna

Tuner at Rig





LGD – AT-1000 ProII



COMET CAT-300

Remote Tuner at Mast Base

Dipole Antenna

450 Ω
Ladder-line
(30 - 50 ft.)



•LDG RT-100 Watt Remote Mount Tuner and RC-100 Control Unit

Auto-Tuner at
Base of mast

Choke Balun

Transceiver

Long coaxial
Cable run
(600 ft. +)



ICOM AH-4

Mobile NVIS Antennas
Loops



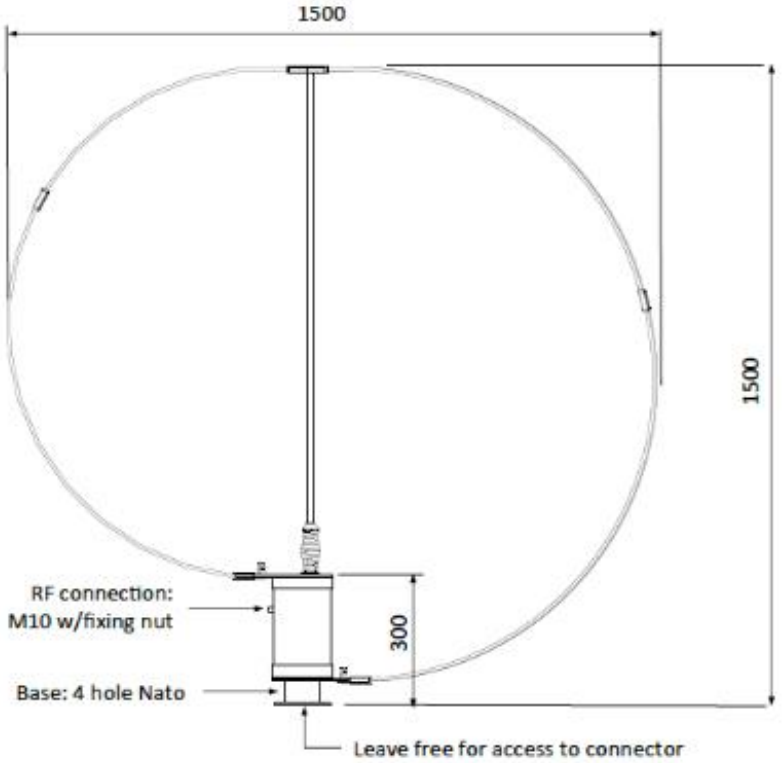


Harris - RF-3134C-AT003/5

TSE®
Tactical Support Equipment
www.tserecon.com

On the Move
Vehicle HF Loop Antenna

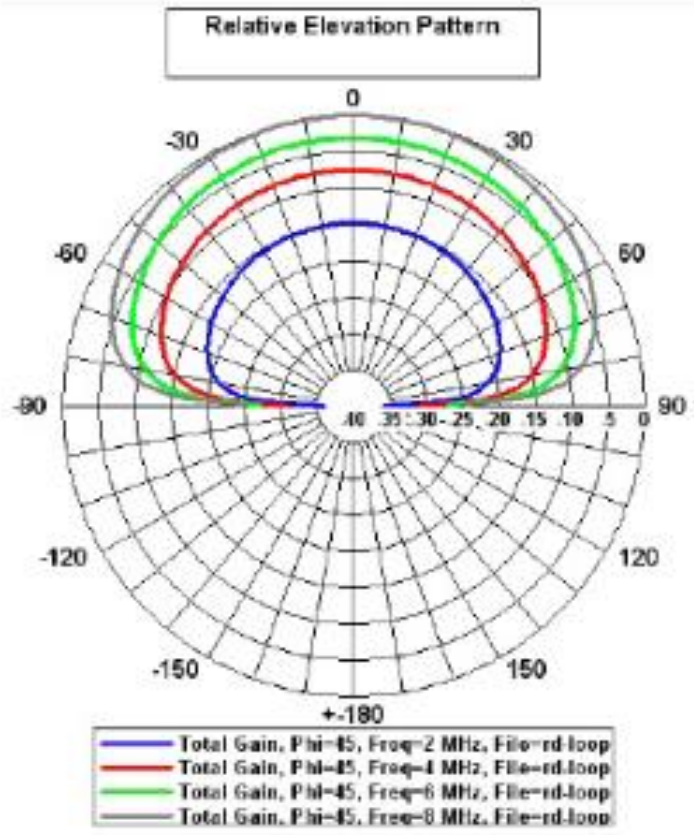


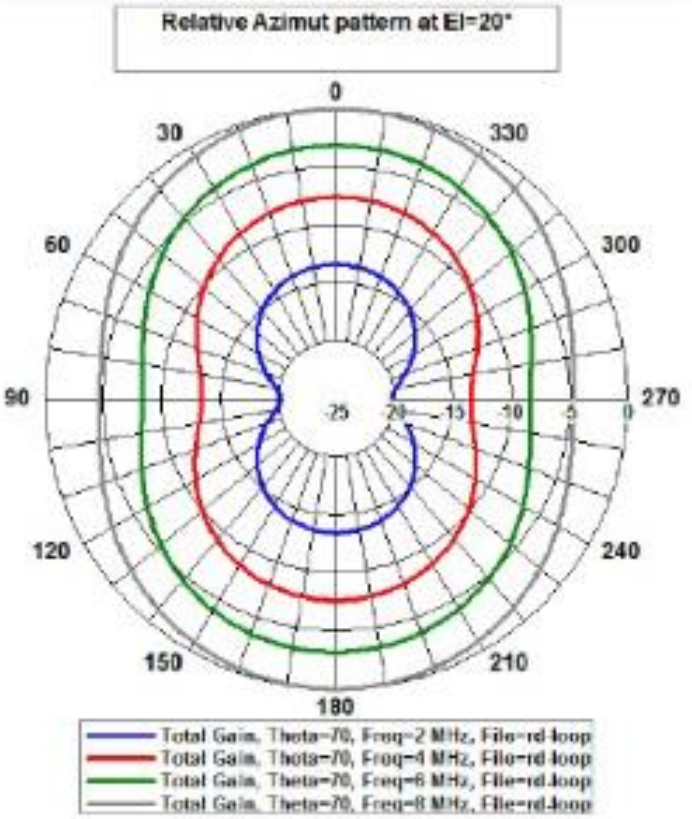


TSE-ANT-VEHHF-OTM

Specifications (TSE-ANT-VEHHF-OTM):	
Electrical	
Frequency	2-30 MHz
Gain	-15dBi @ 3.5MHz, -8dBi @ 10MHz
Impedance	1 to 1000 ohm, 50 ohm nominal
Radiation Pattern	See diagram overleaf
Power Rating	200 Watts PEP and Average

TSE-ANT-VEHHF-OTM





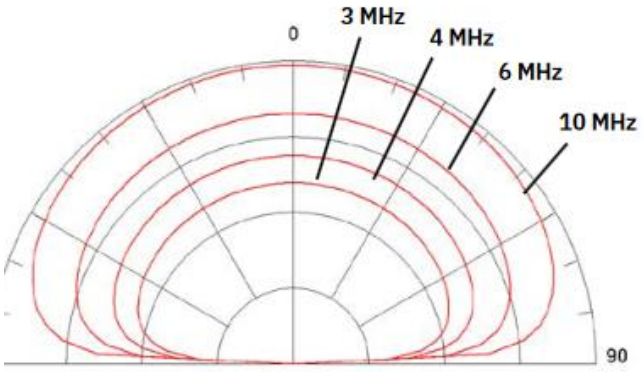
L3HARRIS RF-3134-AT003/5

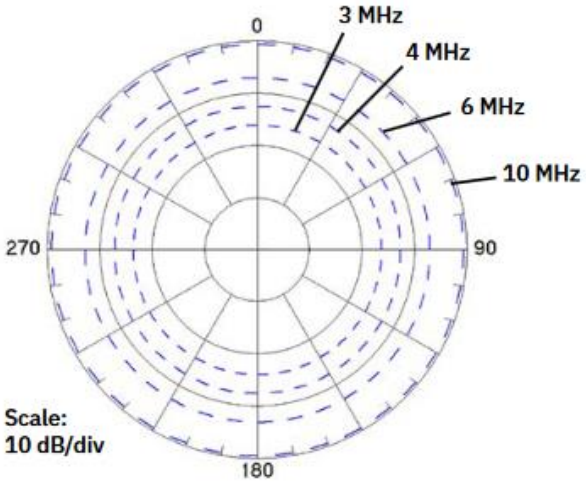




L3HARRIS RF-3134-AT003/5

RF-3134C-AT003/5 Radiation Patterns





ELECTRICAL	
Frequency Range	2-30 MHz
Polarization	Horizontal in the zenith direction
Impedance	50 Ohms (nominal)
VSWR	< 2.0:1, when used with a L3Harris Coupler
Gain	> -24 dBi at 2 MHz, > -15 dBi at 3.5 MHz, > -8 dBi at 10 MHz, > -3 dBi at 30 MHz
Po RF-3134C-AT003/5	150 W Continuous
Radiation Pattern	Quasi hemisphere

ENVIRONMENTAL	
Temperature	-40°F to 140°F (-40°C to +60°C)

MECHANICAL	
Dimensions	59 H x 59 W in (150 H x 150 W cm)
Weight	17.6 lbs (8 kg)
RF Connector	17 mm nut on a 10mm threaded stud
Color	Black matte finish

NVIS Equipment Requirements

- HF Coverage – 2-30 MHz
- Built-in auto-tuner
- Transmit power levels - 100 watts.
- Proper lightning protection between the antenna and rig.



ICOM IC-7300

Operating Modes

- SSB voice
- CW
- Digital Modes
  - WSJT-X modes like FT-8, FT-4
  - PSK-31 (Digipan)
  - RTTY (MMTTY)
  - FLDigi
    - MT-63
    - Olivia
    - MFSK-16
  - Mil Std. 188-110A (MARS)
  - Winlink Radio E-mail using VARA
  - See: 3rd Edition of Get On The Air with HF Digital - ARRL

Summary

- To communicate around Texas and FEMA region 6 we must operate in the NVIS mode of HF propagation.
- This is a highly reliably HF mode. I am on Army MARS Region 6 nets 5 days a week, morning and evening.
- The best way to generate the required NVIS directivity pattern is with a horizontal antenna at a height of less than 0.2 wavelengths.
- The efficiency of the NVIS antenna drops as the height is decreased below 0.2 wavelengths.
- Wide-bandwidth NVIS antennas can have horizontal (azimuth) directivity variance that can be mitigated in a number of ways.
- The SWR of a wide-bandwidth NVIS antenna must be addressed to minimize transmission line losses and maximize power transfer to the antenna.

Questions?

Lewis Thompson – W5IFQ

Personal Cell – 512-587-9944
Home E-mail: w5ifq@att.net